



Maximizing nitrogen removal in a membrane bioreactor (MBR) via a swing-basin and dissolved oxygen reduction configuration: A bio-inspired machine learning optimization approach

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ABSTRACT

Conventional membrane bioreactor (MBR) systems often experience reduced nitrification at low temperatures and limited denitrification under high dissolved oxygen conditions, resulting in insufficient total nitrogen (TN) removal. To overcome these limitations, this study proposes a novel configuration that integrates a swing basin to enhance nitrification during cold periods and a dissolved oxygen reduction (DO-R) tank to improve denitrification by minimizing oxygen interference. Moreover, the complexity of the MBR system necessitated machine learning (ML)-based optimization to enhance TN removal while maintaining effective chemical oxygen demand (COD) and total phosphorus (TP) removal. Long-term operational data (>1 year) from a full-scale wastewater treatment plant in South Korea with a treatment capacity of 25,000 m³/d were analyzed using ML algorithms. The random forest ML model outperformed the deep neural network, achieving a lower RMSE for TN prediction (1.73 vs. 2.81). Shapley Additive exPlanations (SHAP) analysis revealed that the swing basin enhances nitrification by flexibly switching between aerobic and anoxic modes, thereby increasing the effective aerobic volume and hydraulic retention time under low-temperature conditions. In addition, the DO-R tank improved denitrification by lowering DO levels, eliminating competition between oxygen and nitrate. The Harris Hawks bio-inspired optimization algorithm achieved the most accurate convergence, identifying an optimal DO-R tank sludge return flow rate of 146.23 m³/d and yielding effluent COD, TN, and TP concentrations of 5.96, 5.26, and 0.06 mg/L, respectively. Integrating the proposed configuration with ML-driven optimization provides a mechanism-based strategy for improving TN removal, significantly enhancing nitrogen removal in full-scale MBR systems.

1. Introduction

Municipal wastewater treatment plants (WWTPs) integrate a sequence of physical, chemical, and biological processes to progressively remove pollutants from influent streams [1–4]. Among these, biological treatment is essential for the elimination of organic matter, nitrogen, and phosphorus via microbial transformations [5–7]. Over the past two decades, membrane bioreactor (MBR) technology has become an

attractive alternative to conventional activated sludge systems due to its high treatment efficiency, small footprint, and ability to maintain high biomass concentrations [8–11]. Typically, MBRs operate through a combination of anaerobic, anoxic, and aerobic environments that support distinct microbial processes for nutrient and organic pollutant removal [12,13].

However, conventional MBR configurations frequently experience insufficient total nitrogen (TN) removal due to two interrelated

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limitations: (i) low temperatures inhibit nitrifying bacteria activity, reducing nitrification during winter; and (ii) excessive dissolved oxygen (DO) in recycled mixed liquor inhibits denitrifiers, impairing denitrification. These combined constraints severely compromise simultaneous nitrogen removal in real-world wastewater treatment applications [14–18].

Numerous studies have attempted to overcome these shortcomings by modifying MBR configurations or operating conditions. For example, controlling free ammonia concentration was used to enrich nitrification–denitrification (ND) microbial communities, achieving TN removal up to 97.12% with 33% lower carbon consumption [19]. An intermittently aerated moving bed MBR facilitated partial nitrification and denitrification under low chemical oxygen demand (COD)/N conditions, achieving 86.14% TN removal [20]. Other studies integrated elemental sulfur-based autotrophic denitrification with coagulant dosing, enhancing nitrate and phosphorus removal while maintaining stable nitrogen performance [21]. Likewise, MBR systems with pre-denitrification stages or sulfur-based carriers improved TN removal by 30–40% under optimized conditions [22,23].

While previous studies have advanced nitrogen removal in MBR systems, most have focused on optimizing isolated parameters, such as aeration mode, coagulant dosing, or carrier type, rather than on the synergistic integration of complementary strategies. This fragmented approach leaves critical challenges unaddressed. Specifically, low temperatures inhibit nitrification, causing ammonia breakthrough and unstable TN removal [19,24]. Simultaneously, oxygen imbalance between aerobic and anoxic zones suppresses denitrification by exposing denitrifiers to excessive DO [25]. These dual limitations are further compounded by the frequent neglect of nitrogen-phosphorus interactions, particularly under carbon-limited conditions typical of industrial wastewater, where TN removal becomes even more challenging.

To address these challenges, the present study proposes a novel MBR configuration specifically designed to enhance nitrogen removal while maintaining high COD and total phosphorus (TP) removal. Given the influent characteristics of the studied industrial wastewater, nitrogen removal was prioritized in the design. The modified system incorporates two key engineering innovations, including swing basin and DO reduction (DO-R) tank: (i) The swing basin operates between anoxic to aerobic modes, allowing flexible adjustment of aerobic and anoxic volumes to maintain effective nitrification under varying influent and temperature conditions. (ii) The DO-R tank temporarily halts aeration downstream of the aerobic zone, allowing microorganisms to lower DO and maintain the redox balance for efficient denitrification. This research presents a systematic approach for the evaluation and optimization of MBR performance, based on the integration of machine learning (ML) models with bio-inspired optimization algorithms. ML was utilized to model the complex and nonlinear relationships between operational conditions and treatment performance in the MBR system, which are difficult to describe using conventional mechanistic models. Random forest (RF) and deep neural network (DNN) models were developed to predict the simultaneous removal of TN, COD, and TP. Shapley Additive exPlanations (SHAP) analysis was applied to enhance model interpretability and to quantify the contribution of each proposed innovation. Three bio-inspired algorithms were compared with the differential evolution (DE) method to identify optimal operating conditions for maximizing nitrogen removal. While DE is a reliable and widely used optimization method, it typically requires more iterations and longer computation times. In contrast, bio-inspired algorithms offer faster calculation speeds, but their accuracy must be validated against a benchmark method. Therefore, this study compared the convergence results of each bio-inspired algorithm with those obtained from DE to identify which algorithm most closely approximates DE's optimal solution. The algorithm demonstrating the closest agreement with DE, while maintaining high computational efficiency, was selected as the most suitable optimization approach for practical MBR operation.

This work provides a comprehensive process and data-driven

framework that combines engineering modifications, including swing basin and DO-R tank, with ML-based optimization. Unlike previous studies focusing on isolated operational adjustments, this study introduces a synergistic process configuration validated at full scale and interprets its performance using explainable ML, enabling both operational optimization and mechanistic understanding. The proposed configuration enhances biological stability, ensures efficient pollutant removal under varying influent conditions, and contributes novel insights into integrated process design for advanced MBR systems.

2. Materials and methods

2.1. Full-scale MBR plant and data collection

2.1.1. Plant description and operational data source

This study analysed long-term operational data from a full-scale MBR system at the “D” Public Wastewater Treatment Plant in South Korea. The plant has a designed capacity of 25,000 m³ /day, and data were collected over a period exceeding one year. The dataset included daily average values for key water quality parameters (COD, TN, TP) and operational parameters (e.g., flow rates, aeration patterns, and temperatures) from the influent and various process zones. The average influent concentrations were 15.50 mg/L COD, 27.91 mg/L TN, and 1.80 mg/L TP. This comprehensive dataset was used to train and validate the ML models for predicting pollutant removal performance.

2.1.2. Integrated MBR process configuration

The MBR unit primarily serves as a solid-liquid separation device, retaining biomass to ensure a high mixed liquor suspended solids (MLSS) concentration and producing a high-quality effluent. The overall process layout is illustrated in Fig. 1. The system incorporates two key modifications to a conventional MBR unit to address common limitations such as oxygen imbalance and temperature sensitivity: 1) Swing basin for nitrification enhancement. 2) DO-R tank to prevent oxygen intrusion into the anoxic zone. The following sections detail the design and intended function of each unit, providing the process-level context for the operational data analyzed in this study. The primary target of these innovative sections is to enhance TN removal while maintaining a balanced removal of TP and COD.

2.1.3. Process unit design and functional rationale

2.1.3.1. Anaerobic zone. The anaerobic zone serves as the critical starting point for establishing the phosphorus removal process and strategically pre-conditioning the wastewater for subsequent nitrogen removal. Under strictly enforced oxygen-free and nitrate-free conditions, a specialized microbial community known as Phosphate-Accumulating Organisms (PAOs) is selectively enriched [26,27]. These PAOs, unable to use oxygen or nitrate for respiration, instead hydrolyze their internally stored pools of polyphosphate to generate energy. This hydrolysis releases soluble phosphate into the surrounding mixed liquor. This metabolic sequence is fundamental: the release of phosphate here is not a failure but a prerequisite for the subsequent uptake of phosphorus in the aerobic zone. By sequestering a significant portion of the incoming soluble COD as PHA within the PAO cell mass, the anaerobic zone effectively “pre-packages” this carbon source, making it unavailable for ordinary heterotrophic bacteria in the downstream anoxic zone. These forces denitrifying bacteria to rely on more slowly degradable carbon sources, which reduces carbon competition. As a result, energy-rich polyhydroxyalkanoates (PHA) are preserved for the aerobic metabolism of polyphosphate-accumulating organisms (PAOs), supporting their growth and enhancing biological phosphorus removal.

2.1.3.2. Anoxic zone. This zone operates under strictly anoxic conditions, characterized by the absence of DO and the presence of nitrate

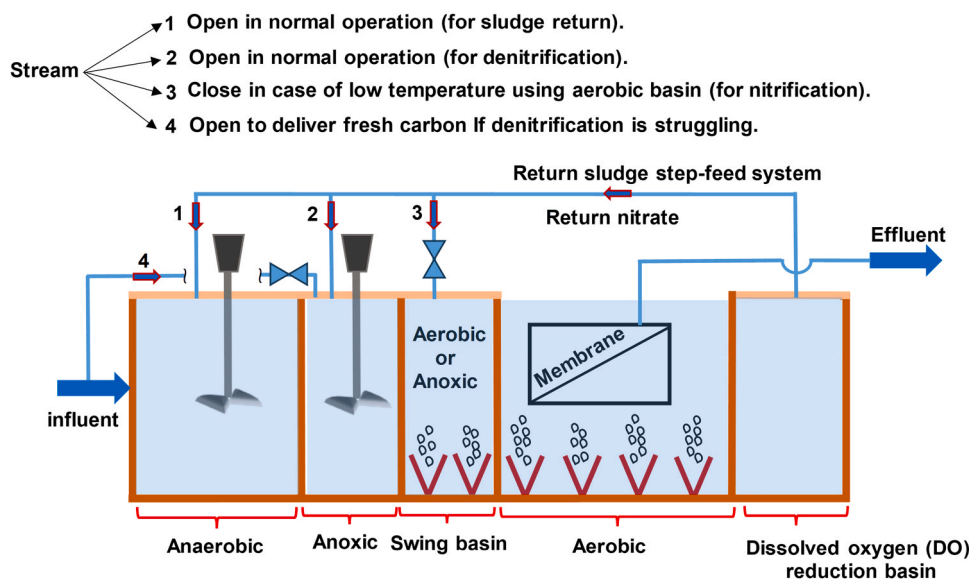


Fig. 1. The configuration of the MBR reactor in this study.

(NO_3^-) and nitrite (NO_2^-). Under these conditions, it serves as the primary site for denitrification, which is the key process governing total nitrogen removal [28,29]. Here, facultative heterotrophic bacteria, which can switch between metabolic pathways, utilize these oxidized nitrogen species as terminal electron acceptors for respiration replacing oxygen. In this biochemical reduction process, nitrate is sequentially transformed into a series of intermediates—nitrite, nitric oxide, and nitrous oxide—before its final conversion to harmless nitrogen gas (N_2), which is liberated into the atmosphere. To ensure the efficiency and completeness of this process, it is paramount to minimize oxygen intrusion from adjacent aerobic zones, as the presence of DO would suppress the enzymes responsible for denitrification, causing the heterotrophic bacteria to preferentially utilize the more energetically favorable oxygen, halting nitrogen removal.

2.1.3.3. Swing basin. The swing basin is a zone with operational flexibility, capable of being switched between anoxic and aerobic modes. This is typically controlled by turning the aeration system on or off. Its primary function is to act as a "buffer" for the nitrification process, which is carried out by autotrophic bacteria that are highly sensitive to low temperatures. In standard systems, low temperatures inhibit nitrification, leading to ammonia accumulation in the effluent and ultimately, the failure of TN removal.

The primary role of the swing basin in TN removal is to ensure stable and complete nitrification, particularly during winter when low temperatures suppress nitrifying activity in conventional aerobic zones. By operating the swing basin in aerobic mode under these conditions, the effective aeration volume and hydraulic retention time are increased, sustaining ammonia oxidation and preventing ammonia accumulation. This operational flexibility allows the swing basin to function as a process buffer, compensating for reduced nitrifier activity and maintaining consistent nitrification performance. The nitrate produced is subsequently recycled to the anoxic zones, where it is reduced to inert N_2 gas through denitrification, enabling high and stable TN removal throughout the year.

The swing basin indirectly enhances COD removal by stabilizing nitrification, which prevents soluble organic nitrogen (e.g., ammonia) from contributing to effluent COD. Effective nitrification also eliminates the need for supplemental carbon addition to treat high ammonia levels, avoiding an increase in organic loading. Moreover, a stable nitrifying community supports the overall mineralization of organic matter, further improving COD reduction. For TP removal, the swing basin

maintains stable nitrification and minimizes oxygen competition between nitrifiers and PAOs, allowing PAOs to efficiently uptake phosphorus in the aerobic zone and thereby enhancing TP removal.

2.1.3.4. Aerobic zone and MBR. In the aerobic reactor, continuous and vigorous aeration creates an environment that supports two critical, parallel biological processes essential for advanced treatment. Firstly, it facilitates nitrification, a two-step aerobic process in which autotrophic bacteria convert ammonium (NH_4^+) to nitrate (NO_3^-) via nitrite (NO_2^-), making the nitrogen available for removal in the anoxic zone [30]. Simultaneously, this oxygen-rich environment fuels the metabolism of PAOs, which use their internally stored PHAs as an energy source to take up soluble phosphate from the mixed liquor at rates far exceeding their metabolic needs. This excess phosphate is then sequestered within their cell mass as polyphosphate, achieving the core objective of enhanced biological phosphorus removal.

2.1.3.5. DO reduction basin. This is a dedicated, non-aerated tank placed after the high-DO environment of the aerobic zone and MBR. The primary and most critical function of the DO-R tank is to enhance denitrification, leading to efficient TN removal. Denitrification is an anoxic process that requires the absence of free oxygen. When oxygen-rich mixed liquor is recycled directly from the aerobic zone/MBR to the anoxic zone, the facultative bacteria responsible for denitrification will preferentially use the free oxygen as an electron acceptor instead of nitrate (NO_3^-) [31–34]. This oxygen poisoning of the anoxic zone severely inhibits denitrification, leading to elevated nitrate levels in the effluent. The DO-R tank directly addresses this by creating an environment for endogenous respiration, where microorganisms consume the residual DO in the absence of an external carbon source. This effectively strips DO from the recycle stream, ensuring that the anoxic zone receives a flow with minimal oxygen concentration. By securing the necessary anoxic conditions, the tank is fundamental to maximizing the denitrification potential of the system.

The DO-R tank is essential for TP removal because it maintains a strictly anaerobic environment, preventing oxygen and nitrate intrusion that would disrupt PAO metabolism and inhibit phosphorus release and carbon uptake. By eliminating DO and nitrate, the tank enables proper initiation of the PAO metabolic cycle, leading to more effective biological phosphorus removal and lower effluent TP. In addition, by enhancing denitrification and PAO activity, the DO-R tank ensures that limited influent carbon is efficiently utilized for anoxic and anaerobic

processes rather than being wasted through undesired aerobic respiration, improving overall COD removal.

2.1.3.6. Return sludge step-feed system. This system provides dynamic control over the internal recycle of nitrate-rich mixed liquor from the aerobic zone. Instead of a single stream, it features separate lines that can send this flow to either the anaerobic zone or the swing/anoxic zone, with adjustable flow rates.

2.1.3.7. Influent bypass (Line 4). The influent bypass is an operational override mechanism designed to combat the common challenge of carbon limitation for denitrification directly. This line strategically diverts a portion of the raw influent, which is rich in readily biodegradable COD such as VFSS, directly to the anoxic zone, bypassing the preceding anaerobic reactor. This feature is explicitly activated when monitoring data indicates a drop in nitrate removal efficiency, signaling that the denitrifying bacteria in the anoxic zone are starved of electron donors.

2.2. Data cleaning and normalization

The dataset was first preprocessed to ensure consistency and usability for ML models. After data cleaning and preprocessing, a total of 2191 observations remained and were used for the ML analysis. All feature and target variables were checked for formatting issues, such as commas or whitespace in numeric fields, which were removed to allow for conversion into floating-point numbers. Any missing or invalid values were imputed using the mean strategy through the SimpleImputer function, thereby preventing data loss while maintaining statistical consistency [35,36]. To improve comparability across features with different scales and units, standardization was applied using the StandardScaler, which transforms the features into zero mean and unit variance. This normalization step ensures that each variable contributes equally to model training and avoids bias towards features with larger numerical ranges.

The operating parameters used in the ML modeling of the MBR system are provided in Table 1. The selection of these 18 feature variables was further justified by their direct relevance to the study's core research objectives. Temperature parameters across the anaerobic, anoxic, and aerobic zones were specifically included to capture seasonal suppression of nitrifying activity, directly addressing the low-temperature nitrification limitation identified as a key research gap [37,38]. Oxidation-reduction potential (ORP) and DO measurements across all stages were designated to quantify the redox conditions governing denitrification efficiency and PAO metabolism, thereby enabling investigation of the oxygen imbalance problem [39,40]. Mixed liquor volatile suspended solids (MLVSS) were incorporated alongside MLSS to distinguish active microbial biomass from inert solids, providing a more accurate indicator of the biological potential for nutrient removal. Sludge volume index (SVI) was included to monitor sludge settleability, as poor settling can lead to biomass washout and deteriorated effluent quality in MBR systems [41–43]. The return sludge flow rate (Sludge-R) was specifically selected because it controls the recycle of nitrate-rich mixed liquor to the anoxic zone, directly influencing denitrification capacity. By systematically including parameters that capture both environmental conditions and biological state variables, the feature set ensures that the ML models learn the underlying process mechanisms rather than merely fitting correlations, enhancing model interpretability and enabling meaningful SHAP analysis of each innovation's contribution.

2.3. Implementation of machine learning algorithms

Two types of ML approaches were employed: Random Forest Regressor (RFR) and DNN. A RFR is an ensemble ML algorithm that predicts continuous values by combining the outputs of many decision

Table 1

Full names, units, and abbreviations of the machine learning parameters used in this study.

Type	Stage	Full Name	Unit	Abbreviation
Feature	Anaerobic	Temperature in the anaerobic tank	°C	Tem-An
		pH in the anaerobic tank	–	pH-An
		Mixed liquor suspended solids in the anaerobic tank	mg/L	MLSS-An
		Dissolved oxygen in the anaerobic tank	mg/L	DO-An
		Oxidation–reduction potential in the anaerobic tank	mV	ORP-An
	Anoxic	Temperature in the swing (anoxic) basin	°C	Tem-Ax
		pH in the swing basin	–	pH-Ax
		Mixed liquor suspended solids in the swing basin	mg/L	MLSS-Ax
		Dissolved oxygen in the swing basin	mg/L	DO-Ax
		Oxidation–reduction potential in the swing basin	mV	ORP-Ax
	Aerobic	Temperature in the aerobic basin	°C	Tem-Ae
		pH in the aerobic basin	–	pH-Ae
		Mixed liquor suspended solids in the aerobic basin	mg/L	MLSS-Ae
		Mixed liquor volatile suspended solids in the aerobic basin	mg/L	MLVSS-Ae
		Dissolved oxygen in the aerobic basin	mg/L	DO-Ae
	DO reduction tank	Sludge volume index	ML/g	SVI-Ae
		Dissolved oxygen	mg/L	DO-R
		Return sludge flow	m ³ / d	Sludge-R
Target		Chemical oxygen demand	mg/L	COD
		Total nitrogen	mg/L	TN
		Total phosphorus	mg/L	TP

trees. Each tree is trained on a random subset of the data and features, and its predictions are averaged to reduce overfitting and improve accuracy [44]. DNNs are computational models inspired by the structure of the human brain, consisting of layers of interconnected nodes (neurons) that process information [45]. They learn patterns from data by adjusting the weights of connections through training, typically using backpropagation to minimize prediction errors. DNNs can model complex, non-linear relationships, making them powerful tools for regression, classification, and pattern recognition.

Table 2 reveals libraries and modules used in Python. The RF model, implemented using the RandomForestRegressor class from scikit-learn, was chosen for its robustness against overfitting and ability to capture non-linear feature interactions. On the other hand, DNN was implemented using TensorFlow/Keras. The DNN was structured with multiple

Table 2

Libraries and modules employed in Python.

Library	Module / Function
pandas	read_csv, DataFrame, to_numeric
numpy	np.sqrt, np.array, np.inf
scikit-learn	train_test_split, cross_val_score, RandomizedSearchCV, RandomForestRegressor, StandardScaler, SimpleImputer, r2_score, mean_squared_error, mean_absolute_error
matplotlib	pyplot (plt)
seaborn	histplot
tensorflow / keras	Sequential, Dense, Dropout, compile, fit, predict, clone_model

fully connected layers and dropout regularization to capture complex, non-linear patterns in the dataset. This dual-model approach provided a comparative assessment between ensemble-based methods and deep learning techniques for multi-output regression tasks, namely, predicting COD, TN, and TP simultaneously. The most accurate algorithm was selected for subsequent analyses, including SHAP interpretation and the optimization process.

2.4. Hyperparameter optimization

Hyperparameter optimization was conducted to enhance model performance and prevent underfitting or overfitting. The RF model's hyperparameters were tuned using a grid search coupled with cross-validation [46]. The optimized hyperparameters for the RF model were set as: `n_estimators=500`, `min_samples_split=2`, `min_samples_leaf=1`, `max_features='log2'`, `max_depth=None`, `random_state=42`, and `n_jobs=-1`. For the DNN, manual hyperparameter tuning was performed by testing different combinations of hidden layer sizes and dropout rates. Models were trained for multiple epochs with batch gradient descent, and the optimal configuration was selected based on validation loss. This systematic tuning ensured that both ML frameworks operated under optimized settings for fair evaluation. The final tuned architecture of the DNN algorithm is detailed in Table 3.

2.5. Model performance evaluation

The root mean square error (RMSE) values for the 5-fold cross-validation data were calculated to evaluate the model accuracy. In cross-validation, the dataset is systematically partitioned into several subsets (or “folds”), and the model is trained on a portion of the data while being tested on the remaining unseen portion [47]. This process is repeated multiple times so that every subset serves as a test set once, and the overall performance is averaged across all folds. Such an approach provides a reliable and unbiased assessment by minimizing the risk of overfitting and ensuring that the model generalizes well to independent data. This makes cross-validation a more effective and robust method compared to a single train–test split. Furthermore, the MAE and MSE values were also calculated to provide additional validation of the model performance, ensuring consistency across multiple error metrics and offering a more comprehensive evaluation of prediction accuracy.

2.6. Explainable machine learning models

The best-performing model was selected and further analyzed using explainable ML methods, including SHAP and Partial Dependence Plots (PDP), to investigate the interactive effects and combined contributions of the input features [48]. These methods clearly revealed how each experimental variable influenced the target response, enhancing the model's transparency and interpretability. By applying explainable ML, the study benefited from increased trust in the predictions, improved

accountability and reproducibility, identification of potential biases or overfitting, and guidance for optimizing experimental conditions to achieve maximum performance.

2.7. Process optimization

During optimization, the process variables were adjusted to reduce the effluent concentrations of COD, TN, and TP, thereby maximizing the system's overall treatment efficiency. Based on the most accurate ML algorithms, the DE method, as a reliable benchmark, was compared with three recently developed bio-inspired optimization algorithms, including the Slime Mould Algorithm (SMA), the Mayfly Optimization Algorithm (MOA), and the Harris Hawks Optimization (HHO). These algorithms are inspired by distinct biological processes and incorporate different search mechanisms, allowing a comprehensive evaluation of their effectiveness in minimizing multiple targets in complex, nonlinear systems. Fig. 2 illustrates the concepts of the optimization algorithms used in this study, represented through their underlying natural behaviors.

DE is an evolutionary optimization technique that employs mutation, crossover, and selection operators to evolve a population of candidate solutions iteratively. It generates diversity by perturbing individuals with scaled differences of randomly chosen solutions, ensuring robust exploration of the search space [49]. DE is well recognized for its simplicity, low computational cost, and stable convergence behavior, making it one of the most widely adopted classical optimizers. Although its convergence can be slower compared with newer metaheuristics, DE's key strength lies in its robustness and reliability across a broad range of continuous optimization problems, establishing it as a dependable baseline for comparison.

SMA is inspired by the oscillatory foraging behavior of slime mould, where adaptive movements are governed by positive and negative feedback mechanisms [50]. It utilizes bio-oscillation equations to update the positions of search agents, thereby dynamically balancing exploration and exploitation. By adjusting the search strategy according to the fitness of candidate solutions, SMA demonstrates strong capability in avoiding local minima. It is characterized by global search efficiency, rapid convergence, and competitive optimization accuracy. Owing to its novelty and flexibility, SMA is particularly suitable for tackling high-dimensional and complex engineering or environmental optimization problems.

MOA mimics the swarming and mating behavior of mayflies, integrating attraction, mating, and flight dynamics into its search strategy [51]. The algorithm divides the population into male and female mayflies, with males performing attractive flights toward superior solutions and females attracted to males for mating. Offspring are generated through recombination processes, maintaining population diversity and preventing premature convergence. MOA achieves a balanced trade-off between exploration and exploitation, resulting in stable convergence performance. Its principal advantages include flexibility, high accuracy, and applicability to multi-objective and nonlinear optimization problems.

HHO is inspired by the cooperative hunting strategies of Harris Hawks, particularly their surprise pounce and dynamic prey-escaping mechanisms [52]. The algorithm models different hunting phases, including exploration, transition, and exploitation, based on the escaping energy of the prey, thus adaptively shifting between global and local search modes. This dynamic adaptation enables HHO to effectively avoid local optima and accelerate convergence toward the global optimum. Key features include adaptive search capability, efficient convergence, and robustness in navigating complex multimodal landscapes. HHO has shown particular strength in solving challenging optimization problems where maintaining exploration–exploitation balance is critical.

Table 3

Deep neural network architecture details.

Layer Type	Units (Neurons)	Activation Function	Dropout Rate	Description
Input Layer	18 (features)	–	–	Receives normalized input features
Hidden Layer 1	128 or 64	ReLU	0.2 – 0.3	Fully connected layer with dropout regularization.
Hidden Layer 2	64 or 32	ReLU	0.2 – 0.3	Second dense layer with dropout to reduce overfitting.
Output Layer	3 (targets: COD, TN, TP)	Linear	–	Predicts continuous output values for COD, TN, and TP simultaneously.

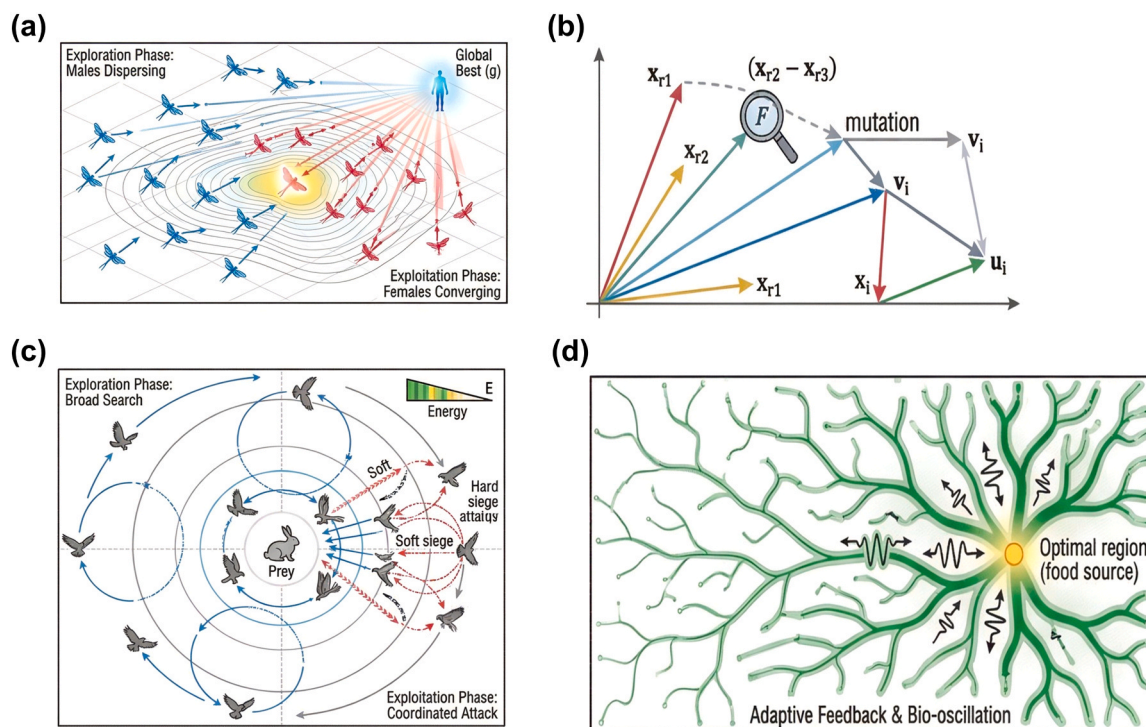


Fig. 2. Conceptual visualization of the optimization algorithms employed in this study, illustrated through their underlying natural behaviors: (a) Mayfly optimization algorithm, (b) differential evolution algorithm, (c) Harris Hawks optimization algorithm, and (d) Slime Mould algorithm.

3. Results and discussion

3.1. Feature analysis of different operational parameters

Fig. 3 presents the ranges of key operational features and their Pearson correlation matrix, offering insight into the interactions among process variables and their influence on treatment performance. The selected parameters vary greatly in scale, including DO-Ax (0.1–0.5 mg/L), ORP-Ax (−400–400 mV), Tem-An (10–24 °C), DO-R (0.05–0.48 mg/L), Sludge-R (19–905 m³/d), and MLVSS-Ae (2000–8000 mg/L). This variability necessitates data normalization or standardization to prevent parameters with larger magnitudes from disproportionately influencing the model. The Pearson correlation coefficient (r) ranges from −1 to +1 values which indicate the direction and strength of the linear relationship between variables. Positive correlations signify direct associations, while negative values reflect inverse trends. Despite their numerically weak values, several correlations hold high process-level significance. This reflects the complex interdependencies inherent in the multistage MBR configuration, which includes a swing basin and a DO-R tank.

Nitrification and denitrification performance are closely governed by redox regulation across the system, as indicated by the positive correlation between TN and oxidation–reduction potential in the anoxic tank (ORP-Ax, $r = 0.31$). This moderate correlation suggests that maintaining stable redox conditions is critical for effective nitrogen transformation, rather than implying direct linear control. The swing basin supports this regulation by flexibly switching between aerobic and anoxic modes, buffering redox fluctuations. Extending the aerobic duration at low temperatures enhances nitrification efficiency by providing sufficient time for ammonia-oxidizing activity [53]. In contrast, dissolved oxygen reduction (DO-R, $r = -0.04$) and sludge return flow (Sludge-R, $r = -0.04$) exhibit weak but negative correlations with TN, reflecting their indirect roles in limiting oxygen intrusion and regulating biomass distribution within the anoxic zone.

A moderate positive correlation between temperature in the anoxic zone (Tem-Ax, $r = 0.30$) and COD removal suggests that temperature-dependent microbial activity enhances organic matter degradation,

with the swing basin indirectly supporting this effect by stabilizing upstream nitrification and redox conditions through flexible aerobic–anoxic operation. In contrast, the weak positive correlation between sludge return from the DO-R basin (Sludge-R, $r = 0.07$) and COD removal indicates limited linear dependence. At the process level, the DO-R basin conditions the return sludge under low-DO conditions to enhance its heterotrophic activity. This preconditioned sludge subsequently improves COD removal efficiency in downstream treatment by utilizing organic matter more effectively while minimizing oxygen interference in anoxic zones [54].

The weak negative correlation between dissolved oxygen in the DO-R basin (DO-R, $r = -0.16$) and TP removal shows a limited linear dependence. A moderate negative correlation between MLSS-Ax ($r = -0.37$) and TP removal indicates that elevated biomass concentrations in the anoxic zone may hinder phosphorus removal efficiency. Residual oxygen in this zone can disrupt the anaerobic–anoxic conditions required for effective biological phosphorus removal. The presence of oxygen suppresses the metabolism of polyphosphate-accumulating organisms, impairing the phosphorus release–uptake cycle essential for efficient removal [55]. The swing basin partially offsets this limitation by flexibly regulating aerobic–anoxic conditions, stabilizing polyphosphate-accumulating organism activity and preventing excessive biomass accumulation in the anoxic zone.

3.2. Comparison of model accuracy

Fig. 4 and Fig. 5 present the actual versus predicted values, residual plots, and residual distributions for the RF and DNN models, respectively. The actual versus predicted values for RF show a narrower and more concentrated distribution of data points around the ideal diagonal compared to DNN, indicating that RF provides more consistent and accurate predictions across the dataset. This improved performance stems from RF's ensemble structure, which leverages multiple decision trees trained on random subsets of features and data. This approach reduces overfitting, stabilizes predictions, and better captures non-linear relationships, particularly with input features of varying scales and

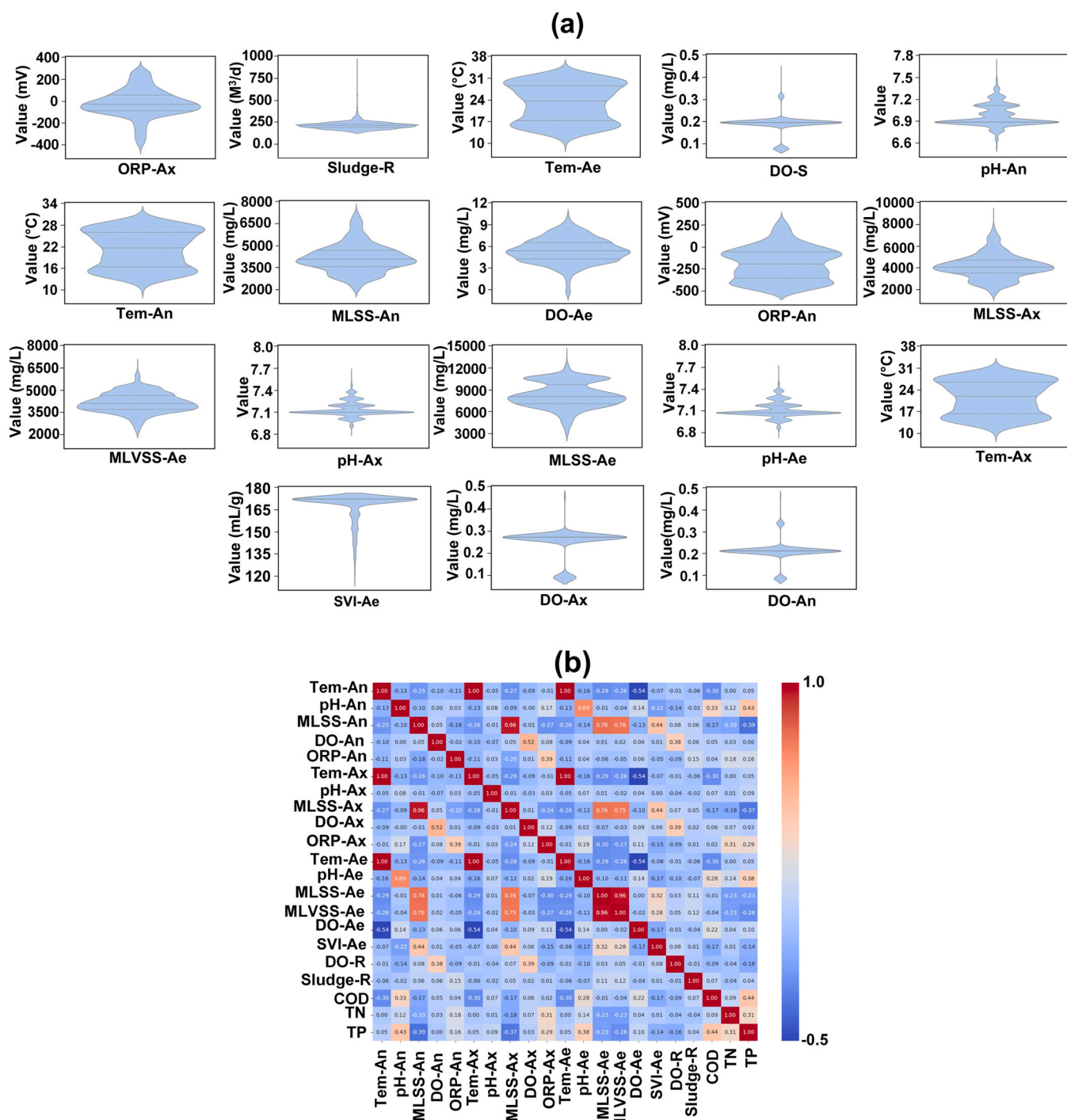


Fig. 3. (a) Feature ranges and (b) Pearson correlation matrix.

distributions.

In the residual plots, RF data points are closer to zero than DNN, reflecting more minor prediction errors and higher reliability. This is likely because RF averages the predictions of multiple trees, mitigating the impact of outliers or extreme values in the inputs. In contrast, DNN can be more sensitive to data sparsity or non-uniform feature distributions. Similarly, the residual distributions for RF exhibit a shorter range and lower spread compared to DNN, indicating reduced variability and fewer large deviations in the predictions. These results demonstrate that RF outperforms DNN, primarily because its ensemble averaging efficiently handles diverse and non-linear input features, stabilizes predictions, and reduces both bias and variance, making it particularly

suitable for modeling complex environmental datasets with multiple interacting parameters.

Fig. 6 presents the RMSE values attained from each fold of the 5-fold cross-validation for each target variable. For RF, the lowest and highest RMSE values are 1.44–1.73 for TN, 0.14–0.17 for TP, and 1.02–1.20 for COD, whereas for DNN, the corresponding ranges are 1.42–2.81 for TN, 0.17–0.31 for TP, and 1.10–2.56 for COD. These results demonstrate that RF not only achieves lower RMSE values overall but also exhibits greater stability across the five folds compared to DNN. Greater stability across the five folds indicates that the RF model performs consistently, regardless of data splitting. This demonstrates strong generalization and low sensitivity to variations in the training set. This consistency suggests

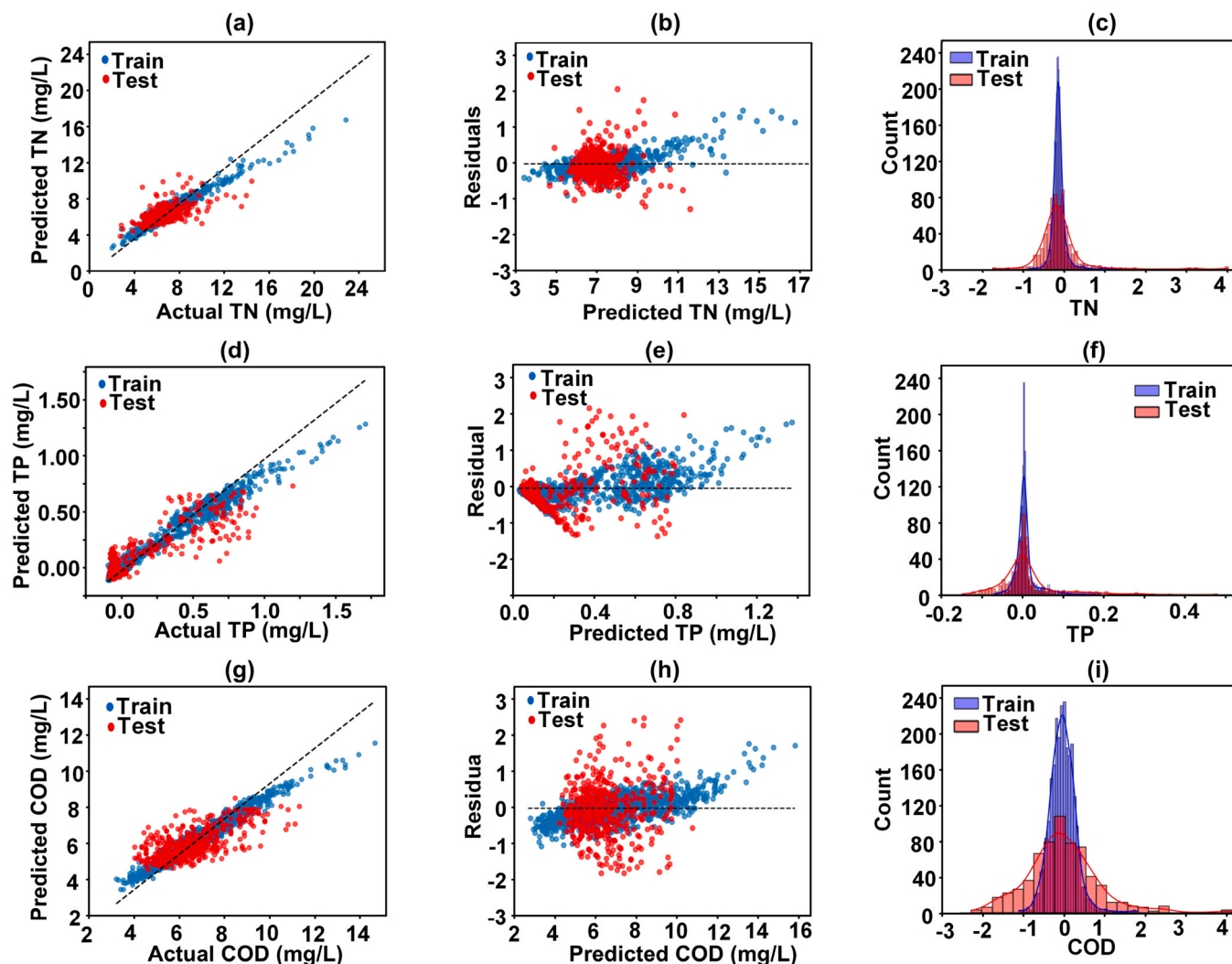


Fig. 4. Actual and predicted values, residual plots, and residual distributions obtained using the RF algorithm: (a–c) TN, (d–f) TP, and (g–i) COD.

that the model is capturing the underlying patterns of the data rather than overfitting to specific subsets. Moreover, the MAE values for the RF model are 0.372, 0.032, and 0.303 for TN, TP, and COD, respectively, with corresponding MSE values of 0.160, 0.370, and 0.300. In comparison, the DNN model yields higher MAE values of 1.038, 0.147, and 0.846 for TN, TP, and COD, respectively, along with corresponding MSE values of 1.399, 4.067, and 0.058. These results reveal that the RF model achieves superior predictive accuracy, robustness, and generalizability across all target variables, making it a more reliable approach for modeling complex environmental processes.

The relationship between model complexity and prediction error is illustrated by plotting RMSE against the number of trees for the RF model and hidden layer size for the DNN model (Fig. 6g–i). In both models, the mean RMSE are shown to evaluate model stability. The RF model exhibits a smoother and more stable trend with smaller variations in RMSE across different numbers of trees, indicating consistent predictive performance [56]. The DNN model shows larger fluctuations in RMSE as the hidden layer size changes, reflecting greater sensitivity to network architecture. Consequently, the RF model achieves lower RMSE values and narrower error bounds than the DNN model, suggesting that it provides more accurate and stable predictions for the studied dataset.

3.3. Explainable machine learning

3.3.1. SHAP analysis

The SHAP interaction ranking plots are provided to illustrate the effects of key operational parameters associated with the swing basin and DO-R tank on TN (Fig. 7a). Among swing-basin-related parameters, anoxic-zone temperature (T_{em-Ax}) is identified as an influential factor affecting nitrification and TN removal. SHAP results indicate that low temperatures exert strong negative contributions to TN removal, particularly under winter conditions, with SHAP values for the TN target ranging from -0.23 to $0.1.05$. However, the magnitude of this negative temperature effect is substantially reduced when swing-basin operation is employed. Extending the aerobic section enhances nitrification by providing sufficient DO and reaction time for ammonia-oxidizing and nitrite-oxidizing bacteria to convert ammonium into nitrate [57,58]. Adequate oxygen availability supports the activity of nitrifying microorganisms and promotes stable nitrification kinetics in biological treatment systems.

DO-R and Sludge-R parameters control denitrification efficiency and, consequently, overall TN removal. At high DO-R values, SHAP contributions approach zero, indicating stabilized total nitrogen removal. Denitrifying bacteria preferentially utilize oxygen as an electron acceptor when it is present, which suppresses the biological reduction of nitrate to nitrogen gas [59,60]. Therefore, elevated residual oxygen inhibits denitrification activity and reduces overall nitrogen removal

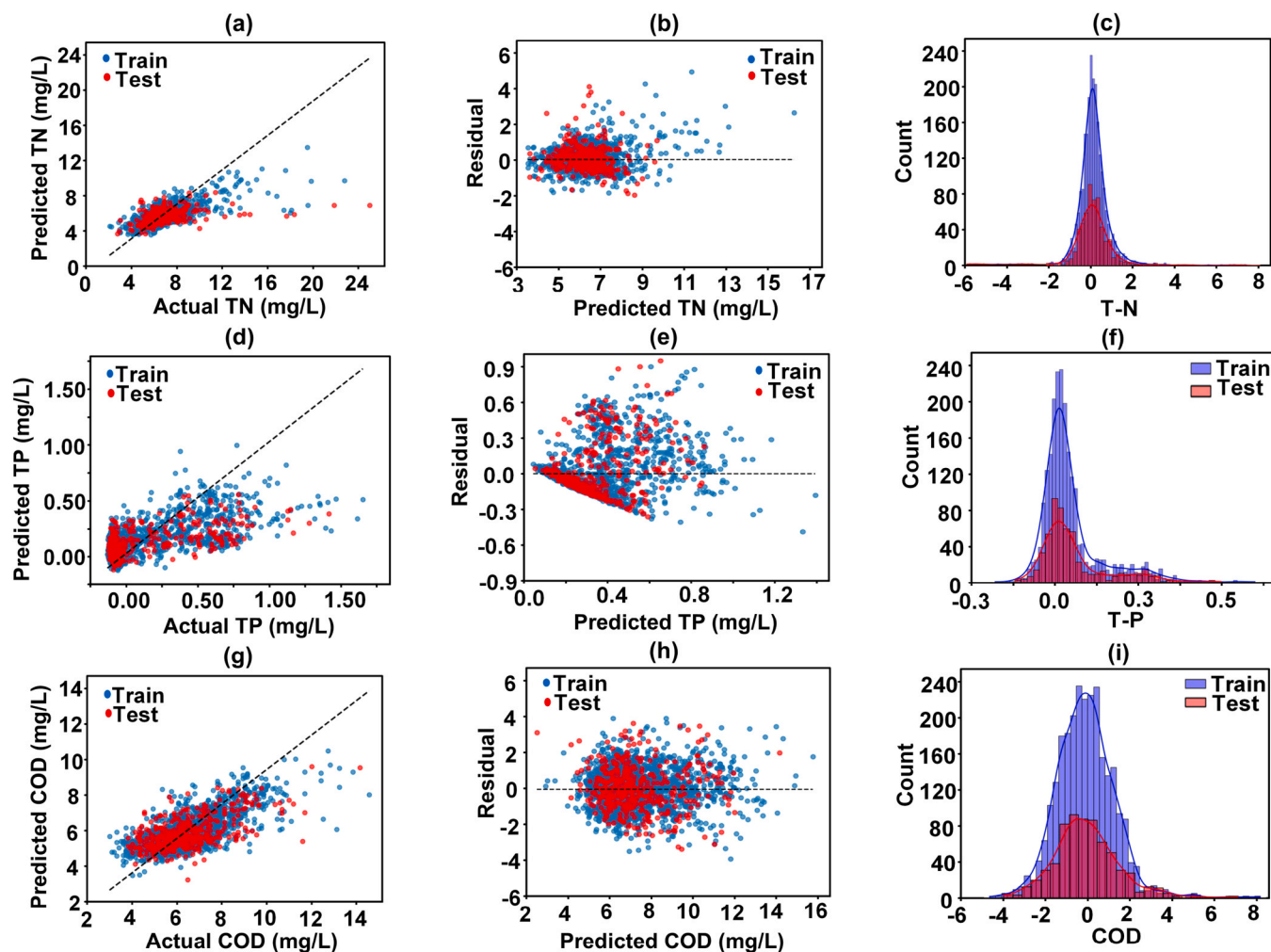


Fig. 5. Actual and predicted values, residual plots, and residual distributions obtained using the DNN algorithm: (a–c) TN, (d–f) TP, and (g–i) COD.

efficiency in the system. The DO-R tank effectively mitigates this limitation by lowering dissolved oxygen levels in the return sludge prior to its re-entry into the anoxic zone, restoring favorable conditions for denitrifiers. This targeted oxygen regulation ensures a smooth redox transition between aerobic and anoxic environments, resulting in improved net TN removal efficiency.

Beyond nitrogen removal, swing-basin operation also influences TP reduction (Fig. 7b). SHAP analysis identifies the oxidation–reduction potential (ORP) in the swing basin as a dominant factor influencing phosphorus removal. Low ORP values (blue points) are associated with negative SHAP contributions, whereas high ORP values (red points) correspond to positive contributions, indicating enhanced phosphorus removal under controlled oxidative conditions. This redox regulation promotes the activity of phosphorus-accumulating organisms and facilitates phosphorus uptake. In addition, the sludge recirculation ratio (Sludge-R) shows an opposing trend, with lower values (blue points) contributing positively to TP removal. In contrast, higher Sludge-R values (red points) change SHAP values toward the negative range, indicating that excessive sludge return adversely affects phosphorus removal. Together, these results highlight that optimized swing-basin redox conditions combined with controlled sludge recirculation are essential for stable and efficient TP removal.

Finally, COD removal is influenced by both DO-R and sludge recirculation parameters, with Sludge-R exerting a stronger impact than DO-R (Fig. 7c). Increasing Sludge-R shifts SHAP values from negative to positive values, revealing COD accumulation due to the recycling of undegraded organic matter and soluble microbial products. In contrast,

higher DO-R values move SHAP contributions toward negative values, reflecting enhanced aerobic oxidation under adequately oxygenated conditions within the DO-R tank. These results highlight that balanced oxygen supply combined with controlled sludge recirculation is essential to prevent organic load accumulation and to sustain effective COD degradation.

The SHAP interaction plots demonstrate that pollutant dynamics are strongly influenced by the coordinated operation of the swing basin and DO-R tank. Nitrogen removal (Fig. 8a–d) shows the most pronounced sensitivity to operational interactions, particularly between DO-R, Sludge-R, and the ORP value of the anoxic and anaerobic tank. At low ORP-Ax values, the DO-R tank minimizes oxygen transfer, sustaining the strongly reducing conditions necessary for complete denitrification. As ORP-Ax increases, DO-R also rises, promoting nitrification in upstream zones but simultaneously restricting denitrification by reducing anoxic time. The swing basin compensates for system imbalances by alternating between aerobic and anoxic phases, a strategy that restores denitrification potential and balances nitrogen conversion efficiency.

Phosphorus removal (Fig. 8e–h) exhibits weaker but still meaningful interactions with DO-R and Sludge-R. At low ORP-Ax in the anoxic basin, both parameters remain low, creating reducing conditions that favor phosphorus release. As ORP-Ax increases under controlled oxygen input from the DO-R tank, phosphorus uptake by PAOs is stimulated. Meanwhile, the swing basin enhances alternating redox conditions, which promote cyclic phosphorus uptake and release dynamics. Interactions between MLSS-An and both DO-R and Sludge-R reveal that higher anaerobic biomass concentrations suppress oxygen transfer and

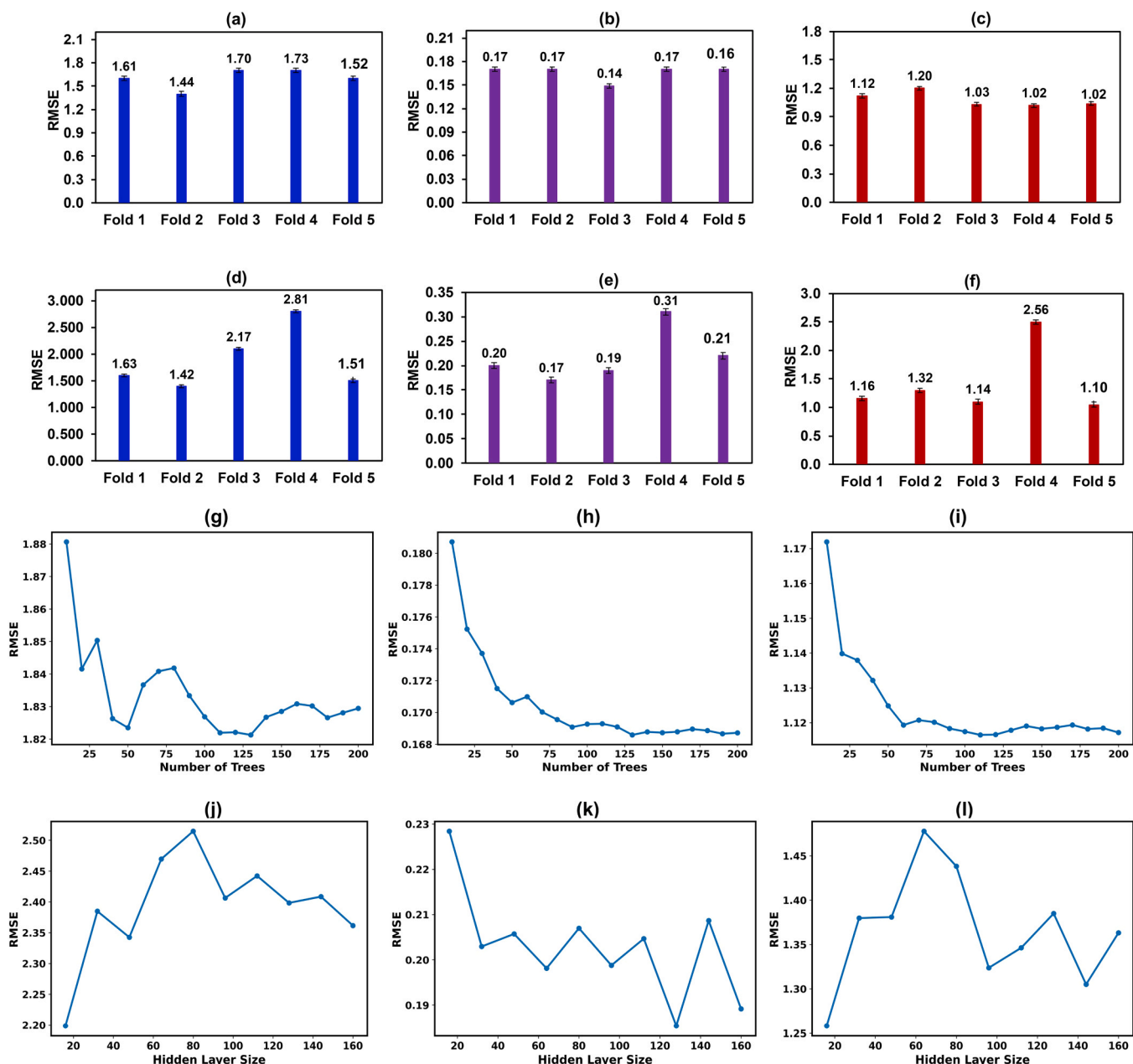


Fig. 6. RMSE values obtained from each fold of the 5-fold cross-validation process: (a) TN, (b) TP, and (c) COD calculated using RF. (d) TN, (e) TP, and (f) COD calculated using DNN. Mean RMSE for the RF algorithm: (g) TN, (h) TP, and (i) COD; and for the DNN model: (j) TN, (k) TP, and (l) COD.

reduce sludge recirculation needs. Higher MLSS levels promote denser sludge flocs and stronger microbial aggregation, which can limit oxygen diffusion into the biomass matrix and maintain more stable anoxic or anaerobic microenvironments. As a result, the reduced oxygen penetration helps maintain appropriate redox conditions for biological processes and decreases the need for intensive sludge recirculation to control dissolved oxygen distribution in the system [61]. These conditions subsequently enhance denitrification by maintaining low DO levels and minimizing nitrate backflow.

For COD (Fig. 8i–l), interactions between pH-An, DO-R, and Sludge-R show that COD variations are governed mainly by oxygen leakage and biomass circulation effects. At elevated anaerobic pH, higher DO-R values increase SHAP contributions for COD, implying that oxygen intrusion from the return sludge disrupts anoxic conditions, weakens denitrification, and allows residual organics to persist. However, when DO is reduced through the DO-R basin, oxygen backflow to the settling

zone is minimized, stabilizing COD degradation efficiency. For the swing basin, SHAP results indicate that temperature is a key factor during winter, with predominantly red points showing that high values strongly influence COD reduction despite reduced microbial activity at low temperatures. By extending effective reaction time and stabilizing operating conditions, the swing basin allows high MLSS to be used more efficiently, leading to improved COD removal under cold conditions.

3.3.2. PDP analysis

The PDP results demonstrate that the integrated operation of the swing basin and DO-R tank strongly governs pollutant removal, with nitrogen dynamics exhibiting the highest sensitivity to operational interactions (Fig. 9a–c). For nitrification, swing-basin control of biomass activity is reflected in the response of aerobic MLVSS. The limited expansion of the high-performance (yellow) region at elevated MLVSS indicates that excessive biomass does not proportionally enhance

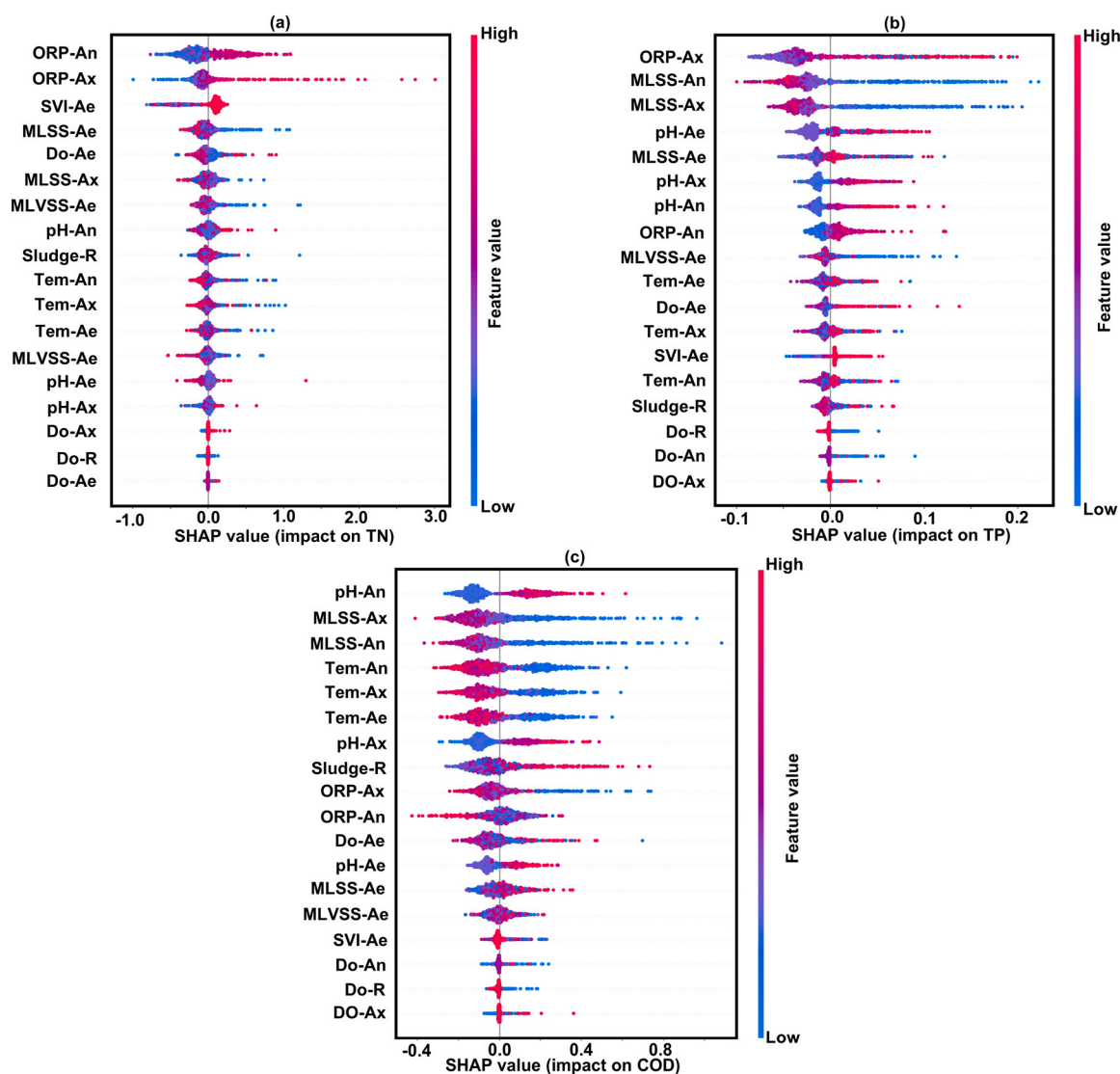


Fig. 7. Interaction ranking plot between features and targets, including (a) TN, (b) TP, and (c) COD.

ammonia oxidation, likely because of oxygen limitation and mass-transfer constraints. In contrast, denitrification and overall TN removal are primarily regulated by Sludge-R and DO-R, with maximum performance observed at high Sludge-R and moderate-to-high DO-R. Elevated DO-R enhances nitrification by promoting ammonium oxidation, while oxygen reduction in the DO-R tank prevents denitrification inhibition. Concurrently, increased Sludge-R supplies nitrate-rich sludge and active biomass to the anoxic zone, sustaining sequential nitrification–denitrification. The swing basin further supports nitrogen removal by enabling dynamic redox transitions, reducing nitrite accumulation, and facilitating nitrogen gas release.

Phosphorus removal (Fig. 9d–f) exhibits secondary sensitivity to operational parameters. Limited yellow regions appear under high Sludge-R and specific ORP-Ax or pH-Ae conditions, indicating that phosphorus removal benefits modestly from increased sludge recycling and controlled oxygen levels. Elevated aerobic pH promotes chemical precipitation with divalent cations, while higher sludge recirculation improves biomass-mediated uptake and transport of phosphorus to aerobic or settling zones. The swing basin indirectly supports TP removal by maintaining redox cycles that allow PAOs to function efficiently.

For COD (Fig. 9g–i), the plots reveal that low anaerobic temperature (Tem-An) and moderate mixed liquor suspended solids (MLSS-An) favor

higher COD removal, reflecting enhanced microbial activity under stable anaerobic conditions. At appropriate MLSS levels, sufficient active microorganisms are present to effectively degrade organic matter, improving the biological oxidation of COD [62,63]. Excessively high MLSS may reduce treatment efficiency due to increased inorganic suspended solids and mass transfer limitations, which can hinder microbial activity and decrease the proportion of active biomass in the system.

Elevated anaerobic pH (pH-An) also contributes positively, as it promotes hydrolysis and fermentation of organic matter, supporting carbon availability for downstream denitrification. The Sludge-R versus DO-R plot shows that COD benefits moderately from increased sludge recirculation and DO in the DO-R tank, but only within a narrow range; excessive DO or sludge return can disrupt redox balance, slightly reducing organic matter degradation. The swing basin's intermittent aerobic–anoxic operation further stabilizes oxygen distribution, preventing organic matter shock loads and supporting steady COD oxidation.

3.3.3. Clarifying the performance impacts of swing-basin and DO-R configurations

The performance of the novel components developed in this study, including the swing basin and the DO-R tank, was evaluated to assess their contribution to total nitrogen removal. Fig. 10a illustrates the

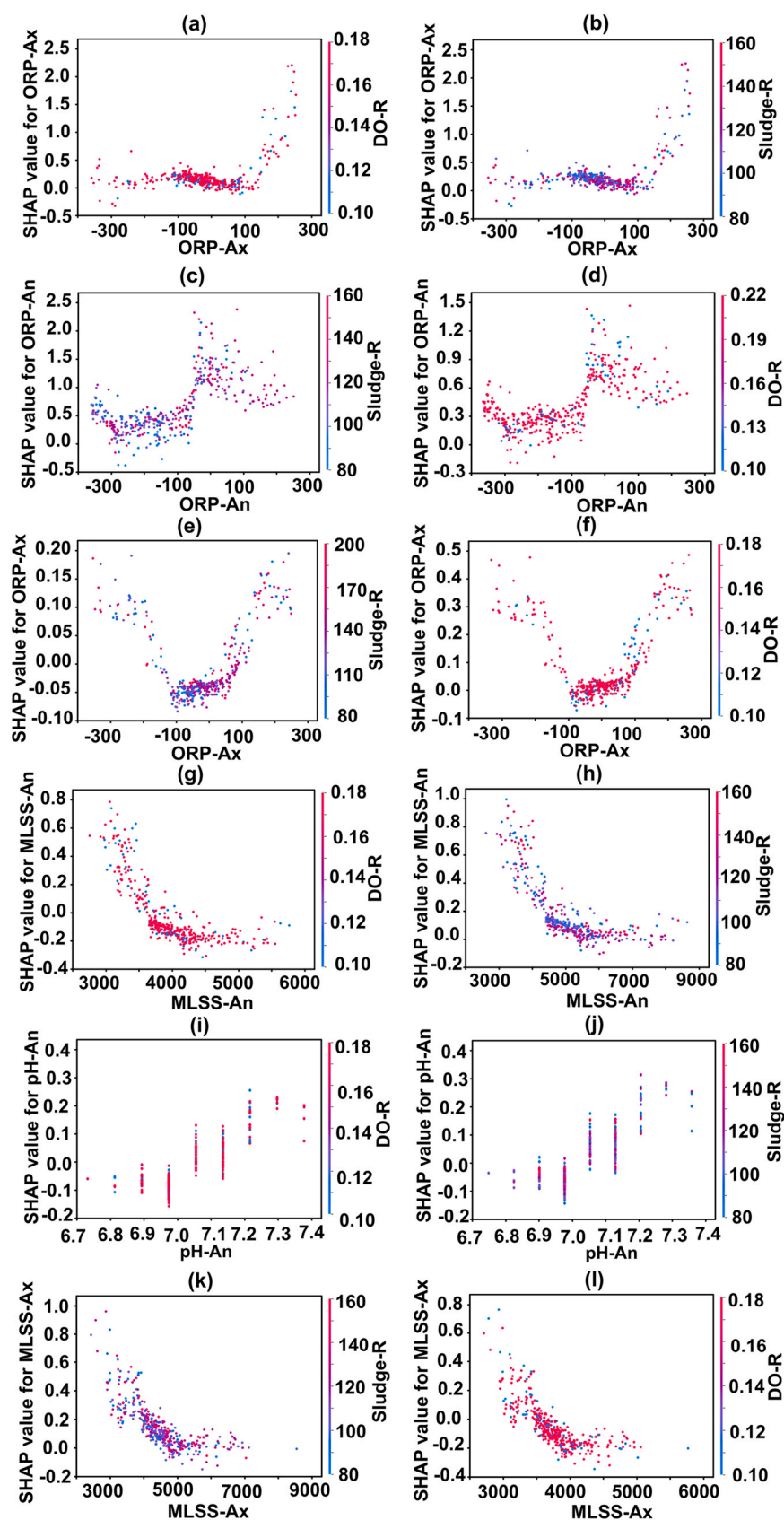


Fig. 8. SHAP interaction plots showing the relationships between important parameters and targets, including (a–d) TN, (e–h) TP, and (i–l) COD.

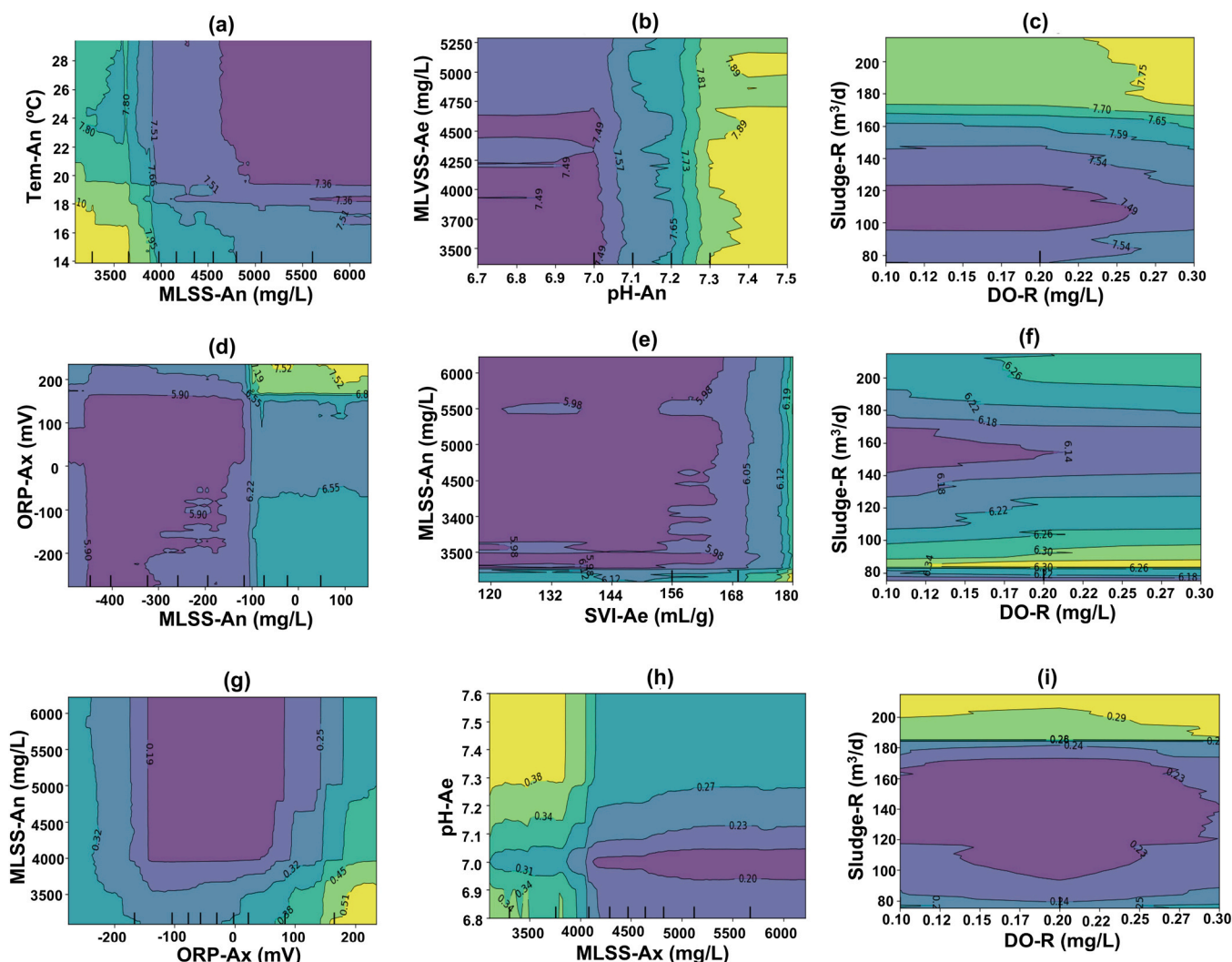


Fig. 9. PDP analysis for (a-c) TN, (d-f) TP, and (g-i) COD.

performance of the swing basin under winter and summer conditions for the conventional configuration, as well as under winter conditions for the proposed configuration. The figure also shows the corresponding effects on TN removal and the influence of key operational parameters.

The swing basin configuration enhances nitrification by dynamically extending the aerobic zone in winter, thereby compensating for temperature-induced inhibition of nitrifying bacteria. At low temperatures, the metabolic activity and growth rate of ammonia-oxidizing and nitrite-oxidizing bacteria decline, leading to reduced nitrification efficiency. The swing basin mitigates this limitation by extending the aerobic zone, which increases oxygen availability and retention time, allowing nitrifiers to sustain ammonia oxidation under cold conditions. This prolonged aerobic exposure stabilizes nitrifier populations and maintains the sequential conversion of ammonium to nitrate, preserving overall nitrogen transformation efficiency. Consequently, the system sustains efficient sequential nitrification–denitrification processes, leading to improved total nitrogen removal across seasonal variations.

In this study, the DO-R tank plays a critical role in enhancing denitrification by creating favorable anoxic conditions (Fig. 10b). In the absence of an external carbon source, microorganisms initially consume the residual dissolved oxygen through endogenous respiration, leading to a progressive decrease in DO concentration. This oxygen depletion suppresses aerobic metabolic pathways and promotes the establishment of an anoxic environment. Since denitrification is an anoxic process that requires the absence of free oxygen, the reduced DO level allows

denitrifying microorganisms to utilize nitrate (NO_3^-) as an alternative electron acceptor. Consequently, higher nitrate availability combined with lower oxygen concentration accelerates denitrification kinetics, resulting in improved total nitrogen removal efficiency.

3.4. Process optimization

Table 4 reveals the optimized feature values and minimized target values calculated using different optimization methods. The DE method, used as the reference, optimized COD to 5.59 mg/L, TN to 5.06 mg/L, and TP to 0.06 mg/L. Key operational parameters include a sludge return flow (Sludge-R) of 139.67 m^3/d and DO-R of 0.20 mg/L. These results indicate that DE provides a balanced configuration of sludge recirculation and oxygen levels that minimizes effluent pollutant concentrations while maintaining process stability. Its optimized parameters serve as a benchmark to evaluate the performance of other bio-inspired optimization algorithms.

The SMA method achieved slightly higher COD (6.16 mg/L) and TN (5.35 mg/L), with TP marginally lower at 0.05 mg/L. Its Sludge-R (138.17 m^3/d) and DO-R (0.20 mg/L) are very close to DE, suggesting that SMA identifies near-optimal operational settings. The minor deviations in COD and TN likely reflect slight differences in biomass distribution and oxygen management compared to DE. Similarly, the MOA algorithm produced slightly higher COD (6.19 mg/L) and TN (5.55 mg/L), with TP at 0.05 mg/L. While the absolute differences in effluent

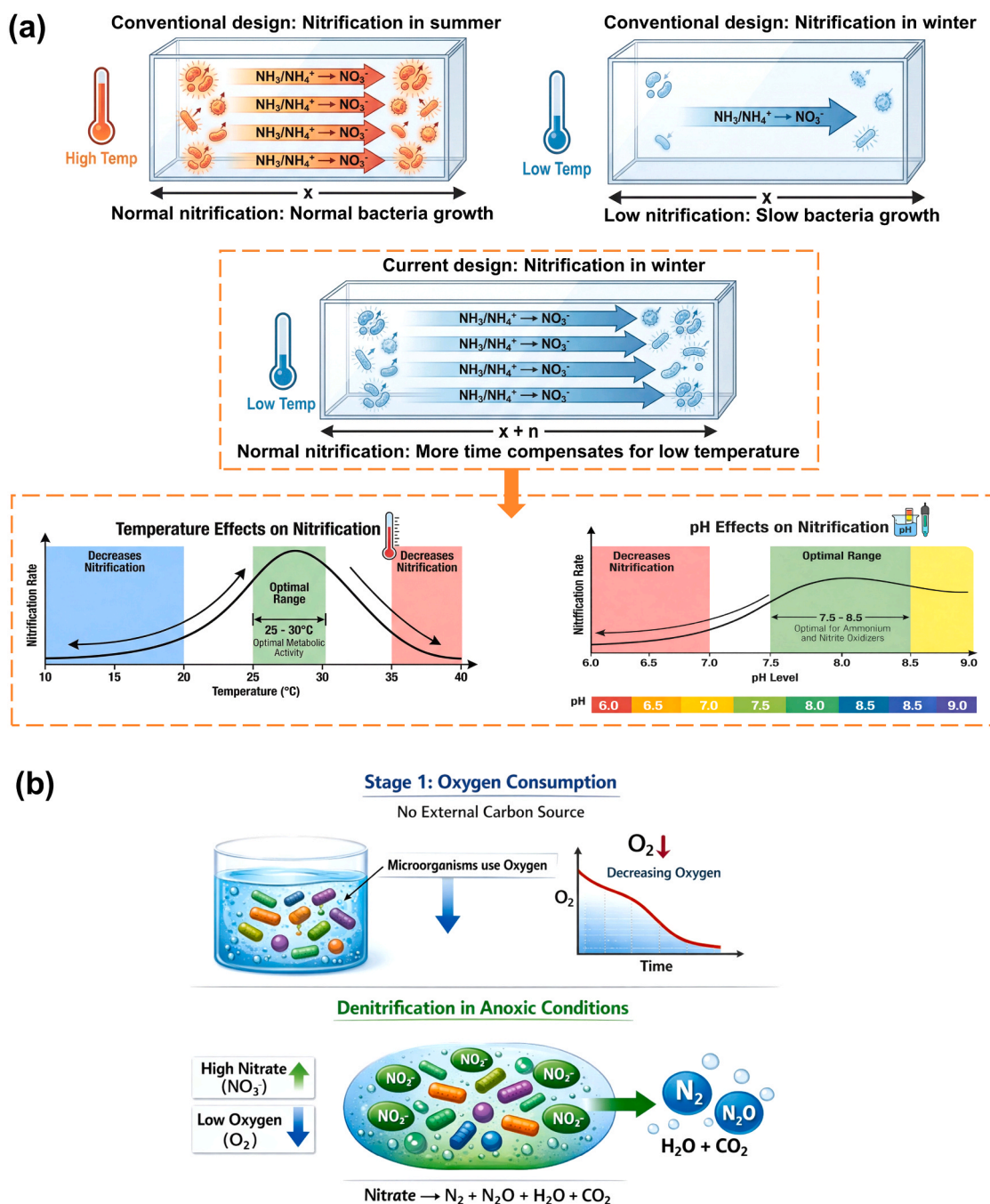


Fig. 10. (a) Comparison between conventional nitrification designs and the swing-basin configuration (this study), highlighting the effects of pH and temperature on the nitrification process. (b) Effect of the DO-R tank on denitrification.

concentration are small (e.g., ~ 0.5 mg/L difference in TN between DE and MOA), their practical significance can be substantial. In the context of increasingly stringent wastewater discharge regulations, a 0.5 mg/L margin can be the difference between consistent compliance and periodic violations. Notably, MOA's Sludge-R ($119.88 \text{ m}^3/\text{d}$) is significantly lower than DE, indicating less sludge recirculation, which may reduce denitrification efficiency and contribute to the higher effluent COD and TN values.

Among the alternative methods, the HHO method yields results closest to DE, with COD at 5.96 mg/L, TN at 5.26 mg/L, and TP at 0.06 mg/L. Its Sludge-R ($146.23 \text{ m}^3/\text{d}$) is slightly higher than DE, while DO-R remains the same at 0.20 mg/L. This suggests that HHO effectively balances sludge return and oxygen control, maintaining efficient

denitrification and phosphorus removal. Among the evaluated algorithms, HHO provided solutions most consistent with the DE benchmark, indicating reliable convergence for this specific MBR optimization problem. The SMA follows, while the MOA shows the largest deviations, particularly in Sludge-R and its subsequent impact on COD and TN.

The HHO predicts results closer to DE by efficiently balancing exploration and exploitation through its unique, energy-driven search strategy [64–66]. In HHO, the transition between exploration and exploitation is dynamically controlled by the prey's escape energy factor E , which decreases over iterations, allowing the algorithm to initially explore the search space broadly and later intensify exploitation around promising solutions. This mimics DE's reliance on differential mutation and crossover to maintain diversity while refining solutions.

Table 4

Optimized feature values and minimized target values obtained using different optimization methods.

Dataset type	Abbreviation	Unit	Differential evolution (DE)	Slime Mould Algorithm (SMA)	Mayfly Optimization Algorithm (MOA)	Harris Hawks Optimization (HHO)
Feature	Tem-An	°C	29	29	20	29
	pH-An	–	7.30	7.03	7.035	6.97
	MLSS-An	mg/L	4804.97	4908.49	4630.98	6126.58
	DO-An	mg/L	0.20	0.20	0.20	0.20
	ORP-An	mV	184.46	265.43	165.61	257.32
	Tem-S	°C	29.71	29.81	20.06	29.79
	pH-S	–	7.01	7.18	7.10	6.95
	MLSS-S	mg/L	4912.48	5262.66	4873.97	5034.03
	DO-S	mg/L	0.20	0.20	0.20	0.20
	ORP-S	mV	-1.71	64.63	123.22	131.01
	Tem-Ae	°C	29.91	26.44	23.69	29.82
	pH-Ae	–	7.00	7.00	7.00	7.00
	MLSS-Ae	mg/L	8035.04	8012.74	7459.47	7968.15
	MLVSS-Ae	mg/L	4958.33	5185.49	4671.50	5072.03
	DO-Ae	mg/L	3.60	6.57	4.81	4.01
	SVI-Ae	ML/g	124.45	123.15	133.42	125.50
	DO-R	mg/L	0.20	0.20	0.20	0.20
	Sludge-R	m ³ / d	139.67	138.17	119.88	146.23
Target	COD	mg/L	5.59	6.16	6.19	5.96
	TN	mg/L	5.06	5.35	5.55	5.26
	TP	mg/L	0.06	0.05	0.05	0.06

Specifically, HHO employs multiple attacking strategies, such as soft and hard sieges combined with progressive rapid dives, which adapt according to the prey's energy and escape probability. This adaptive behavior enables the algorithm to avoid local optima and converge toward global optima, similar to how DE balances mutation and recombination to generate high-quality offspring [67]. This adaptive mechanism enables HHO to effectively explore new regions while exploiting previously identified promising areas, resulting in optimization performance comparable to that of DE. Consequently, it can identify optimal settings for key parameters such as Sludge-R and DO-R, which critically influence the removal efficiencies of COD, TN, and TP. Furthermore, because the HHO algorithm provides fast computational performance while producing results closely comparable to those of DE a reliable but slower optimization method, HHO can be considered an efficient and competitive optimization approach.

4. Conclusion

This study demonstrates that integrating a swing basin with a DO-R tank provides an effective strategy for improving TN removal in full-scale MBR systems while maintaining stable COD and TP removal. The swing basin improved nitrification by extending aerobic conditions, while the DO-R tank enhanced denitrification by reducing residual DO in recycled sludge. The ML analysis of long-term operational data showed that the RF model outperformed the DNN algorithm in predicting COD, TN, and TP concentrations. The HHO optimization algorithm produced results closest to those of DE among the evaluated methods. This performance was achieved at a Sludge-R value of 146.23 m³ / d and a DO-R value of 0.20 mg/L. Beyond the specific case study, this work reveals how integrating process engineering innovations with explainable ML and bio-inspired optimization can provide a general framework for understanding, predicting, and improving the performance of complex biological wastewater treatment systems. One limitation of this study is that the results are based on operational data from a single full-scale wastewater treatment plant, which may limit the generalizability of the ML models and optimization outcomes to systems with different influent characteristics or operational conditions. Future research should evaluate the proposed framework across multiple full-scale plants to confirm its applicability under different operational and influent conditions. Nevertheless, the proposed framework provides practical guidance for improving nutrient removal and offers a scalable strategy for optimizing MBR operation under variable

environmental conditions.

CRediT authorship contribution statement

Euntae Yang: Writing – review & editing, Validation, Methodology, Investigation. **Moon-Hyun Hwang:** Writing – review & editing, Validation, Investigation, Conceptualization. **Sangyoun Lee:** Writing – review & editing, Validation, Investigation, Conceptualization. **Kyu-Jung Chae:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization. **Jung-Min Lee:** Writing – review & editing, Visualization, Validation, Investigation. **Mohammad Khajavian:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] Y. Hu, R. Wei, K. Yu, Z. Liu, Q. Zhou, H. Zhang, M. Zhang, C. Wang, L. Zhang, B. Xue, G. Wang, G. Liu, S. Qu, Evaluating the environmental benefits and resource loads of municipal wastewater treatment plants in China, *Environ. Impact Assess. Rev.* 119 (2026) 108334, <https://doi.org/10.1016/j.eiar.2026.108334>.

- [2] H.E. Al-Hazmi, G.K. Hassan, M. Maktabifard, D. Grubba, J. Majtacz, J. Makinia, Unsupervised and supervised machine-learning modeling of the operational parameters enriching the anammox microbiome in a full-scale municipal wastewater treatment plant, *J. Water Process Eng.* 83 (2026) 109665, <https://doi.org/10.1016/j.envres.2022.114432>.
- [3] Y. Zhao, D. Yu, Y. Wei, Occurrence and transformation of antimony in a full-scale municipal wastewater treatment plant, *J. Hazard. Mater. Lett.* 7 (2026) 100173, <https://doi.org/10.1016/j.hazl.2025.100173>.
- [4] H. Han, Y. Zhao, X. Wu, H. Yang, Intelligent feature extraction for anaerobic/anoxic/aerobic process in municipal wastewater treatment plant, *J. Environ. Sci.* 161 (2026) 360–371, <https://doi.org/10.1016/j.jes.2025.07.014>.
- [5] H.-J. Yan, Y.-W. Cui, H.-K. Liang, Z.-Y. Li, Critical review on magnetic biological effects of microorganisms in the field of wastewater treatment: theory and application, *J. Environ. Chem. Eng.* 13 (2025) 118351, <https://doi.org/10.1016/J.JECE.2025.118351>.
- [6] C. Meyer, M.L. Bär, J. Hollender, Wastewater as a dual indicator of human and environmental exposure to synthetic antioxidants: occurrence and fate in biological and advanced wastewater treatment, *Environ. Int.* 203 (2025) 109709, <https://doi.org/10.1016/J.ENVINT.2025.109709>.
- [7] M.A. Hasanath, A. Panigrahi, N.K. Chaitanya, P. Chatterjee, Treatment of beverage wastewater using microalgae: a focus on COD removal, biomass production, and lipid accumulation, *Bioresour. Technol. Rep.* 33 (2026) 102467, <https://doi.org/10.1016/j.biteb.2025.102467>.
- [8] S. Zaman, M.I. Nelson, M.T. Moores, F.I. Hai, A critical review of the mechanisms of virus removal by membrane bioreactors—Influencing factors and correlation with operating parameters, *J. Environ. Manag.* 380 (2025) 124945, <https://doi.org/10.1016/J.JENVMAN.2025.124945>.
- [9] B. Wu, J. Zhong, H. Chang, D. Li, C. Wang, H. Han, Q. Gu, Y. Zhong, X. Dai, X. Chai, Micro-granular sludge driven by powder carrier in membrane bioreactor: highly-efficient N & P removal and membrane fouling alleviation, *Water Res.* 288 (2026) 124635, <https://doi.org/10.1016/J.WATRES.2025.124635>.
- [10] C. Huang, H. Zhang, J. Huang, L. Zhang, S. Tong, F. Wang, G. Liu, X. Yu, H. Xu, W. Liu, Performance of anoxic and aerobic flocs in anoxic/aerobic-membrane bioreactor (A/O-MBR) for the treatment of printing and dyeing wastewater (PDW), *Biochem. Eng. J.* 225 (2026) 109943, <https://doi.org/10.1016/J.BEJ.2025.109943>.
- [11] Y. Na, L. Ranieri, A. Di Cesare, R. Sabatino, J.S. Vrouwenvelder, L. Fortunato, Evaluating the removal efficiency of emerging contaminants in a gravity-driven membrane bioreactors GD-MBR under various aeration conditions for primary wastewater treatment, *Case Stud. Chem. Environ. Eng.* 10 (2024) 100885, <https://doi.org/10.1016/J.CSCEE.2024.100885>.
- [12] M.E. Fuller, P.C. Hedman, K.H. Chu, T.S. Webster, P.B. Hatzinger, Evaluation of a sequential anaerobic-aerobic membrane bioreactor system for treatment of traditional and insensitive munitions constituents, *Chemosphere* 340 (2023) 139887, <https://doi.org/10.1016/J.CHEMOSPHERE.2023.139887>.
- [13] J. Guo, S. Qiu, L. Dai, L. Zhang, L. Meng, M. Liu, H. Yao, The occurrence and removal of steroid estrogens in a full-scale anaerobic/anoxic/aerobic-membrane bioreactor process and the implication of the bacterial community dynamics, *J. Environ. Chem. Eng.* 10 (2022) 107294, <https://doi.org/10.1016/J.JECE.2022.107294>.
- [14] Z.S. Ma, H.M. Zhang, W.J. Ma, N.Y. Liu, X.C. Yu, Advanced total nitrogen removal in low C/N wastewater through simultaneous nitrification-heterotrophic denitrification coupled with sulfur-driven autotrophic denitrification: performance and microbial succession, *Environ. Res.* 286 (2025) 122839, <https://doi.org/10.1016/J.ENVRES.2025.122839>.
- [15] L. Wu, Y. Tian, Y. Liu, J. Xu, C. Wang, L. Zhang, M. He, Y. Peng, Enhancement of simultaneous partial nitrification-anammox and denitrification nitrogen removal and biofilm formation by novel biochar/nano zerovalent iron modified polyurethane biocarrier for treatment of landfill leachate, *Chem. Eng. J.* 511 (2025) 161958, <https://doi.org/10.1016/J.CEJ.2025.161958>.
- [16] W. Li, X. Jiang, J. Li, J. Li, Simultaneous aerobic nitrogen and phosphorus removal characteristics and mechanism by heterotrophic nitrification aerobic denitrification microorganisms: a review, *J. Water Process Eng.* 76 (2025) 108235, <https://doi.org/10.1016/J.JWPE.2025.108235>.
- [17] M. Ghasemi, S. Chang, S. Sivaloganathan, Modeling and simulation study of simultaneous nitrification-denitrification in membrane aerated bioreactor, *J. Memb. Sci.* 668 (2023) 121210, <https://doi.org/10.1016/J.MEMSCI.2022.121210>.
- [18] Y. Jin, J. Ding, W. Zhan, J. Du, G. Wang, J. Pang, N. Ren, S. Yang, Effect of dissolved oxygen concentration on performance and mechanism of simultaneous nitrification and denitrification in integrated fixed-film activated sludge sequencing batch reactors, *Bioresour. Technol.* 387 (2023) 129616, <https://doi.org/10.1016/J.BIORTECH.2023.129616>.
- [19] Y. Zhang, X. Fang, J. Ye, L. Huang, C. Zhong, Effect of free ammonia control strategy on partial nitrification-denitrification in MBR reactor under low C/N conditions: performance, SMP and EPS, and microbial community, *J. Environ. Chem. Eng.* 13 (2025) 116273, <https://doi.org/10.1016/J.JECE.2025.116273>.
- [20] J. Ye, L. Huang, Y. Zhang, C. Zhong, The impact of low COD/N ratio on denitrification performance and microbial community in an intermittent aeration moving bed membrane bioreactor, *Process Biochem.* 150 (2025) 189–201, <https://doi.org/10.1016/J.PROCBIO.2025.01.012>.
- [21] M. Pyo, D. Kim, H.S. Kim, M.H. Hwang, S. Lee, E.J. Lee, Sulfur powder utilization and denitrification efficiency in an elemental sulfur-based membrane bioreactor with coagulant addition, *Water Res.* 272 (2025) 122882, <https://doi.org/10.1016/J.WATRES.2024.122882>.
- [22] Y.C. Woo, J.J. Lee, A. Jeong, J. Song, Y. Choi, H.S. Kim, Removal of nitrogen by a sulfur-based carrier with powdered activated carbon (PAC) for denitrification in membrane bioreactor (MBR), *J. Water Process Eng.* 34 (2020) 101149, <https://doi.org/10.1016/J.JWPE.2020.101149>.
- [23] M.E. Matsubara, K. Helwig, C. Hunter, J. Roberts, E.L. Subtil, L.H.G. Coelho, Amoxicillin removal by pre-denitrification membrane bioreactor (A/O-MBR): performance evaluation, degradation by-products, and antibiotic resistant bacteria, *Ecotoxicol. Environ. Saf.* 192 (2020) 110258, <https://doi.org/10.1016/J.ECOENV.2020.110258>.
- [24] H.E. Al-Hazmi, X. Lu, D. Grubba, J. Majtacz, M. Badawi, J. Makinia, Metabolic basis and ecological stability of bioaugmented AO-MBR treating low-temperature toilet wastewater: linking functional genes, pathway efficiency, and microbial dynamics, *Process Saf. Environ. Prot.* 209 (2026) 108595, <https://doi.org/10.1016/j.scitotenv.2023.161633>.
- [25] U.B. Antu, M.S. Islam, S. Ahmed, M. Arifuzzaman, S. Saha, P.R. Mitu, A.R. Sarkar, N.A. Mahiddin, Z. Ismail, K.A. Ibrahim, A.M. Idris, An innovative MABR-MBR integrated sequential anoxic-oxic biological (A2/O2) system for treating high-ammoniacal nitrogen containing wastewater, *J. Water Process Eng.* 82 (2026) 109553, <https://doi.org/10.1016/j.jwpe.2024.106317>.
- [26] Q. Guan, C. Jiang, Y. Li, Q. Meng, J. Zhang, Y. Li, Biodegradation of cresyl diphenyl phosphate using microbial consortium enriched from anaerobic activated sludge, *J. Environ. Chem. Eng.* 13 (2025) 119610, <https://doi.org/10.1016/J.JECE.2025.119610>.
- [27] Z. Zhong, W. Ye, B. Li, N.A. Al-Dhabi, J. Zhao, S. Li, Y. Sun, H. Zhang, W. Tang, S. Chen, Phosphate-iron modified Enteromorpha prolifera hydrochar enhances dry anaerobic digestion of food waste: synergistic mechanisms of electron transfer network, microbial consortia remodeling, and metagenomic insights, *Environ. Res.* 289 (2026) 123385, <https://doi.org/10.1016/J.ENVRES.2025.123385>.
- [28] Y. Sui, Y.W. Cui, H.J. Yan, R.C. Yang, High enrichment of heterotrophic nitrification-aerobic denitrification bacterial community achieved by shortening anoxic feeding duration in the sequencing batch reactor: feeding mode effect and mechanism, *J. Environ. Manag.* 390 (2025) 126250, <https://doi.org/10.1016/J.JENVMAN.2025.126250>.
- [29] L. Sun, Y. Liu, H. Shu, Y. Shao, Z. Xu, J. Qi, Development and simulation-based optimization of a Split-flow low-aerobic/anoxic/Oxic process integrating partial nitrification -denitrification with step feeding for enhanced nitrogen removal, *J. Water Process Eng.* 81 (2026) 109322, <https://doi.org/10.1016/J.JWPE.2025.109322>.
- [30] J. Tang, Y. Chen, H. Huang, A. Zhang, Enhanced performance of a semi-aerobic aged refuse biofilter for treating landfill leachate MBR effluent pre-oxidized by a microwave-activated peroxyacetic acid process, *J. Environ. Chem. Eng.* 13 (2025) 117107, <https://doi.org/10.1016/J.JECE.2025.117107>.
- [31] C. Teng, K. Zhou, C. Peng, W. Chen, Ecological mechanisms underlying microbial heterogeneity, influent impact, and pathogen control in bioreactors and supernatant under two-stage anoxic/oxic process upgrade, *Chem. Eng. J.* 529 (2026) 172446, <https://doi.org/10.1016/j.watres.2021.117525>.
- [32] H. Amanollahi, G. Moussavi, S. Giannakis, Enhanced sulfide-driven autotrophic denitrification/ammonium oxidation process via metabolic cooperation of anaerobic bacterial consortia in a packed-bed bioreactor, *Chem. Eng. J. Adv.* 25 (2026) 101062, <https://doi.org/10.1016/j.cjce.2026.101062>.
- [33] T.T.T. Hang, V.V. Phat, H.H. Hanh, D.Van Tri, N.T. Anh, N.T. Nhan, T.Le Luu, Pilot study on an integrated anaerobic membrane bioreactor (AnMBR) and intermittent cycle extended aeration system (ICEAS) for seafood processing wastewater treatment, *Bioresour. Technol.* 445 (2026) 134105, <https://doi.org/10.1016/j.biortech.2026.134105>.
- [34] S. Krishnan, A. Thongsai, P. (Lek) Noophan, M.A. Siddiqui, Y.S. Wong, J. Bae, S. Chairapat, Comparative evaluation of a hybrid membrane treatment train versus full-scale aerobic/anoxic activated sludge system: C-N removal performances, syntrophic microbial dynamics, and energy savings, *Biochem. Eng. J.* 227 (2026) 110052, <https://doi.org/10.1016/j.bej.2025.110052>.
- [35] M.M. Perez, E. Cabrera, C. Rial, Y. You, Y. Wang, K. Weinberger, D.V. Nydam, J. O. Giordano, Screening and selection of a machine learning algorithm for development of a model to select cows for clinical examination using data from automated health monitoring technologies and other predictors of cow health, *J. Dairy Sci.* (2025), <https://doi.org/10.3168/JDS.2025-26511>.
- [36] E. Kim, J.J. Park, G. Lee, J.S. Cho, S.K. Park, D.Y. Yun, K.J. Park, J.H. Lim, Innovative strategies for protein content determination in dried laver (*Porphyra* spp.): evaluation of preprocessing methods and machine learning algorithms through short-wave infrared imaging, *Food Chem. X* 23 (2024) 101763, <https://doi.org/10.1016/J.FOCHX.2024.101763>.
- [37] D. Cudjoe, B. Zhu, H. Wang, A new Anti-inhibition strategy to overcome high ammonia stress for anaerobic co-digestion of food waste and sewage sludge by applying a domesticated anaerobic electrocatalytic cathodic-membrane bioreactor (AnEC-MBR), *Chem. Eng. J.* 527 (2026) 171377, <https://doi.org/10.1016/j.jclepro.2022.132161>.
- [38] C. Chen, M. Sun, Z. Liu, J. Zhang, K. Xiao, X. Zhang, G. Song, J. Chang, G. Liu, H. Wang, X. Huang, Robustness of granular activated carbon-synergized anaerobic membrane bioreactor for pilot-scale application over a wide seasonal temperature change, *Water Res.* 189 (2021) 116552, <https://doi.org/10.1016/j.watres.2020.116552>.
- [39] X. Li, H. Yang, X. Wang, S. Lu, Y. Wang, B. Liu, Y. Zhang, H. Zhao, Z. Tian, X. Zheng, Balancing sludge reduction and membrane fouling mitigation by tuning electrical voltages of a side-flow electrochemical oxidation system during MBR processing, *J. Clean. Prod.* 425 (2023) 138712, <https://doi.org/10.1016/j.jclepro.2023.138712>.

- [40] U. Sathya, Keerthi, M. Nithya, N. Balasubramanian, Evaluation of advanced oxidation processes (AOPs) integrated membrane bioreactor (MBR) for the real textile wastewater treatment, *J. Environ. Manag.* 246 (2019) 768–775, <https://doi.org/10.1016/j.jenvman.2019.06.039>.
- [41] A.A. Albalasmeh, A.A. Berhe, T.A. Ghezzehei, Effects of aeration intensity and sludge return rate on microbial structure change and pharmaceutical removal performance of the novel solar-powered two-stage inclined-plate-membrane bioreactor system for the treatment of real building wastewater, *Chemosphere* 385 (2025) 144603, <https://doi.org/10.1016/j.carbpol.2013.04.072>.
- [42] A. Mineo, A. Cosenza, H.Y. Ng, G. Mannina, Volatile fatty acids from sewage sludge by anaerobic membrane bioreactors: lesson learned from two-year experiments with fouling analysis by the resistance in series model, *Results Eng.* 21 (2024) 101839, <https://doi.org/10.1016/j.rineng.2024.101839>.
- [43] V. Acevedo Alonso, T. Kaiser, R. Babist, T. Fundneider, S. Lackner, The crucial role of activated sludge models (ASMs) on wastewater treatment processes: Developments, applications, and future perspectives, *Water Res.* X 30 (2026) 100497, <https://doi.org/10.1016/j.watres.2021.117079>.
- [44] L. Hu, Y. Peng, H. Yan, F. Qiu, Z. Cheng, A random forest-optimized sensor fusion approach for non-invasive ammonia measurement: enhancing performance in jet impact-negative pressure reactors, *Measurement* 258 (2026) 119121, <https://doi.org/10.1016/J.MEASUREMENT.2025.119121>.
- [45] M. Khajavian, A. Haseli, Modeling the adsorption of ibuprofen on the Zn-decorated S,P,B co-doped C2N nanosheet: machine learning and central composite design approaches, *J. Ind. Eng. Chem.* 137 (2024) 583–592, <https://doi.org/10.1016/J.JIEC.2024.04.002>.
- [46] M.S. Vhatkar, P.S. Mahajan, R.D. Raut, N. Cheikhrouhou, S. Ghoshal, Optimized hyperparameters for retail sales forecasting using grid search, *Eng. Appl. Artif. Intell.* 158 (2025) 111472, <https://doi.org/10.1016/J.ENGAPPAI.2025.111472>.
- [47] M.S. Shah, F. Khan, S. Zendeheboudi, A. Mamudu, An integrated model to investigate optimized CO₂ utilization and risk mitigation in offshore hydrocarbon reservoirs, *Geoenergy Sci. Eng.* 234 (2024) 212589, <https://doi.org/10.1016/J.GEOEN.2023.212589>.
- [48] B. Siddique, F. Ahmad, J. Najeeb, S. Naeem, M. Ali, A.Z. Dewidar, H.O. El-ansary, Machine learning assisted designing of Indacenodithiophene (IDT)-based polymers for future application of photoacoustic imaging, *J. Photochem. Photobiol. A Chem.* 447 (2024) 115215, <https://doi.org/10.1016/J.JPHOTOCHEM.2023.115215>.
- [49] J. Li, Z. Zhang, T. Zheng, J. Tang, Z. Lei, S. Gao, Discrete bi-population differential evolution for optimizing complex wind farm layouts in diverse terrains, *Energy* 335 (2025) 137885, <https://doi.org/10.1016/J.ENERGY.2025.137885>.
- [50] I.M. Conteh, G. Njai, A. Conteh, Q. Du, An optimized feature selection using triangle mutation rule and restart strategy in enhanced slime mould algorithm, *Egypt. Inform. J.* 30 (2025) 100709, <https://doi.org/10.1016/J.EIJ.2025.100709>.
- [51] V.G. Saranya, S. Karthik, Bio-Inspired Intelligent Routing in WSN: integrating mayfly optimization and enhanced ant colony optimization for energy-efficient cluster formation and maintenance, *C. - Comput. Model. Eng. Sci.* 141 (2024) 127–150, <https://doi.org/10.32604/CMES.2024.053825>.
- [52] Z. Yuqi, W. Bing, C. Yuquan, Z. Zhen, H. Qiang, Environmental and economic optimization of office building microgrid based on carbon-electricity market and improved Harris Hawks optimization algorithm, *Energy* 332 (2025) 136818, <https://doi.org/10.1016/J.ENERGY.2025.136818>.
- [53] S. Yao, C. Hu, X. Ou, Q. Xiu, H. He, Y. Hou, X. Zhang, T. Ma, J. Gu, W. Xia, Synergistic nitrogen removal by robust heterotrophic nitrification-aerobic denitrification and substantial assimilation in a novel *Zobellella* sp, *Water Res.* 296 (2026) 125652, <https://doi.org/10.1016/j.watres.2026.125652>.
- [54] H. Makizuka, M.J. Rocco, A. Hafuka, K. Kimura, Long-term operation of a high-rate membrane bioreactor for organic matter recovery from municipal wastewater: Role of iron-associated constituents in fouling and implications for stable performance, *Chem. Eng. J.* 532 (2026) 174245, <https://doi.org/10.1016/j.cej.2026.174245>.
- [55] M. Shafiquzzaman, M.S. Rahman, M.M. Hasan, M.S. Azam, A.T. Ahmed, H. Haider, Performance Evaluation of Low-Cost algal ceramic membrane bioreactor for wastewater treatment operating at Non-Aeration mode and varying organic loading rates, *Bioresour. Technol.* 445 (2026) 134061, <https://doi.org/10.1016/j.biortech.2026.134061>.
- [56] K. Rostami, R. Kumar, A.M. Abdalla, M. Sedighi, M. Ghasemi, Random Forest and differential evolution for maximizing microbial fuel cell performance in wastewater remediation, *Biomass Bioenergy* 208 (2026) 108868, <https://doi.org/10.1016/j.biombioe.2025.108868>.
- [57] R.C. Yang, Y.W. Cui, Z.Y. Li, M.T. Li, Y. Sui, Comprehensive evaluation on potentiality and key bacterial community of heterotrophic nitrification and aerobic denitrification in real wastewater treatment plants, *Process Saf. Environ. Prot.* 208 (2026) 108509, <https://doi.org/10.1016/j.psep.2026.108509>.
- [58] L. Britschgi, S. Wei, A. Proesl, E. Morgenroth, N. Derlon, The critical role of flocs in nitrification in full-scale aerobic granular sludge-based WWTP, *Water Res.* 274 (2025) 123021, <https://doi.org/10.1016/j.watres.2024.123021>.
- [59] S. Banerjee, C.A. Kirkby, D. Schmutter, A. Bissett, J.A. Kirkegaard, A.E. Richardson, Oxygen stress drives overcompensation of carbon sources for enhanced polymer denitrification, *Water Res.* 296 (2026) 125631, <https://doi.org/10.1016/j.soilbio.2016.03.017>.
- [60] B. Xu, L. Liu, M. Wang, Z. Fu, P. Xu, J. Ma, Q. He, Effects of dissolved oxygen on the anaerobic/oxic/anoxic simultaneous nitrification, denitrification and phosphorus removal (AOA-SNDPR) process for advanced wastewater treatment and in situ sludge reduction, *J. Environ. Chem. Eng.* 13 (2025) 115258, <https://doi.org/10.1016/j.jece.2024.115258>.
- [61] W. Sohn, J. Jiang, S. Phuntsho, Y. Choden, V.H. Tran, H.K. Shon, Influence of temperature and dissolved oxygen on nitrification in a membrane bioreactor treating urine, *J. Water Process Eng.* 78 (2025) 108721, <https://doi.org/10.1016/j.desal.2023.116663>.
- [62] Z. Liu, Impact of influent quality on the mixed liquor volatile suspended solids/ mixed liquor suspended solids of mixed liquor from treatment plant, *Desalin. Water Treat.* 280 (2022) 52–59, <https://doi.org/10.1016/j.eti.2020.101190>.
- [63] Y. Tian, H. Cheng, Y. He, H. Feng, W. Shi, J. Sun, Y. Li, Y. Zhang, M. Yang, Z. Tian, Enhanced efficacy of anaerobic membrane bioreactor (AnMBR) for treating pharmaceutical wastewater with abundant protein particles, *Chem. Eng. J.* 527 (2026) 171889, <https://doi.org/10.1016/j.cej.2025.171889>.
- [64] S.S. Usagoonkar, D.R. Edla, R. Dharavath, Optimizing deep CNN architecture via hybrid Harris Hawks arithmetic algorithm for EEG meditation classification, *Neuroscience* 598 (2026) 100–110, <https://doi.org/10.1016/j.neuroscience.2026.02.001>.
- [65] M. Batis, Y. Chen, L. Liu, A.A. Heidari, H. Chen, Prey capture enhanced Harris hawks optimizer for wrapper-based feature selection in high-dimensional medical data, *Comput. Methods Prog. Biomed.* 277 (2026) 109237, <https://doi.org/10.1016/j.cmpb.2026.109237>.
- [66] H. Rezk, A. Alahmer, A.G. Olabi, E.T. Sayed, Application of artificial intelligence and red-tailed hawk optimization for boosting biohydrogen production from microalgae, *Int. J. Thermofluids* 24 (2024) 100876, <https://doi.org/10.1016/j.ijft.2024.100876>.
- [67] M.A. Elaziz, A.A. Heidari, H. Fujita, H. Moayedi, Lens-imaging learning Harris hawks optimizer for global optimization and its application to feature selection, *Expert Syst. Appl.* 202 (2022) 117255, <https://doi.org/10.1016/j.asoc.2020.106347>.